TRANSPORT AND TILING BOUNDS FOR THE AFFINE PLANK PROBLEM ON THE TRIANGLE

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Abstract. Bang's affine plank conjecture asserts that planks covering a convex body have relative widths summing to at least 1; it is open already in the plane. For a convex body K and *directions* given by affine maps $u_i : K \to [0,1]$ we introduce the *transport defect* $D_K(u)$ — the least D admitting a probability measure on K with all u_i-marginals $\le D \cdot \mathrm{Leb}$ — so every covering by planks in these directions satisfies $\sum \mathrm{rw} \ge 1/D_K(u)$, and we prove a minimax duality for D_K with exact certificates on both sides. On the triangle we determine exactly when the transport method certifies the sharp constant 1: for three pairwise non-parallel directions a uniform-marginal witness measure exists ($D = 1$) if and only if the mid-level lines concur; for every concurrent *cyclic* triple we exhibit the witness and the unique tight covering with $\sum \mathrm{rw} = 1$ — a two-parameter family of rigid configurations extending the median case — and we classify all concurrent triples admitting a witness. We also determine, partly by an exact-rational computation, the one-plank-per-direction covering constant at one tilted non-concurrent triple (the three-plank bound). Calibrations: the uniform measure gives $D \le d$ on any d-dimensional body, and the d-simplex facets have $D = (d+1)/2$. No result resolves the conjecture for the triangle, let alone the plane.

1. Introduction

A *plank* of width w is the region between two parallel hyperplanes at distance w. For a convex body $K \subset \mathbb{R}^d$ and a plank P with unit normal u, the *relative width* of P is $\mathrm{rw}_K(P) = \mathrm{w}(P)/\,\mathrm{w}_K(u)$, where $\mathrm{w}_K(u) = \max_{x\in K}\langle u, x\rangle - \min_{x\in K}\langle u, x\rangle$. Bang's affine plank conjecture [3, 1] states:

Conjecture 1.1 (Affine plank problem; Bang 1951)**.** If K is covered by planks $P_1, \dots, P_n$, then $\sum_{i=1}^n \mathrm{rw}_K(P_i) \ge 1$.

Ball [2] proved Conjecture 1.1 for centrally symmetric K. For general bodies it is open, and, as emphasised by Bakaev–Yehudayoff [4], it is open *already in the plane* — only the two-plank case is settled there. The best general lower bound is $\sum \mathrm{rw} \ge 2/(1+\sqrt{d})$ [4]; on the equilateral triangle their method gives $\sum \mathrm{rw} \ge 4\sqrt{3} - 6 \approx 0.928$.

Scope. This paper studies the triangle as a concrete body and develops the associated *transport* (witness-measure) and *tiling* methods. It is *not* an attack on Conjecture 1.1 through the Ambrus reduction: that reduction sends a covering by N planks to a simplex of dimension $2N-1$, so the triangle (the 2-simplex) is never a target (Remark 1.2). Our results are new sharp cases for a fixed body, a rigidity theorem, a dimension-agnostic structural characterization, and an obstruction; we prove nothing about the general conjecture. Since the distinctions matter, we spell them out: five statements should be kept apart — (a) bounds for coverings by planks parallel to a *fixed direction family*; (b) existence and characterization of *witness measures*; (c) bounds for *all* coverings of the triangle; (d) Conjecture 1.1 for the triangle; (e) Conjecture 1.1

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in the plane. Every theorem in this paper lives at level (a) or (b); at level (c) we prove nothing beyond the universal $1/d$ of Theorem 2.5, and (d), (e) remain wide open.

Remark 1.2. Ambrus's reduction [1] sends a covering by N planks to a covering of a simplex of dimension $2N-1$ (two points per plank). The target simplices thus have odd dimension growing with N; the 2-simplex (triangle) is never among them. So proving the conjecture for the triangle does *not* imply the planar case, and our results are about the triangle as a concrete body, not a reduction of Conjecture 1.1.

The triangle model and terminology. By affine invariance take $\Delta = \{x \in \mathbb{R}^3 : x_i \geq 0,\ x_1 + x_2 + x_3 = 1\}$. Throughout, a *direction* on a convex body K is a *normalized affine coordinate*: an affine map $u : K \to [0,1]$ that is onto $[0,1]$. (A direction encodes the pencil of parallel hyperplanes $\{u = t\}$, *not* a Euclidean normal vector; this is the natural datum for relative width.) A normalized plank is $P = \{x \in K : u(x) \in I\}$ with u a direction and $I \subset [0,1]$ an interval; then $\mathrm{rw}_K(P) = |I|$. Covering means $K \subset \bigcup_i P_i$. The restriction $I \subset [0,1]$ loses no generality: replacing I by $I \cap [0,1]$ leaves $P \cap K$ unchanged and does not increase $|I|$.

A *witness measure* for directions $u_1, \ldots, u_n$ on K is a probability measure μ on K whose pushforward under every u_i is Lebesgue measure on $[0,1]$ ($u_{i\#}\mu = \mathrm{Leb}[0,1]$) — Gardner's *relative-width measure* [5]. If one exists, then every covering of K by planks parallel to these directions satisfies $\sum \mathrm{rw} \geq 1$, since $\mu(P) = |I|$ for each plank and the plank masses must sum to at least $\mu(K) = 1$. On Δ, with vertices A, B, C, we single out the *cyclic triples*: $u_1 = (\tau_1, 0, 1)$, $u_2 = (0, 1, \tau_2)$, $u_3 = (1, \tau_3, 0)$ with $\tau_i \in (0,1)$, each triple of numbers listing the values at (A, B, C) (Figure 2); the medians are the case $\tau = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$.

The transport defect as organizing frame. The paper is organized around one quantity. The *transport defect* $D_K(u_1, \ldots, u_n)$ (Definition 2.1) is the least $D \geq 1$ such that some probability measure on K has all its u_i-marginals bounded by $D \cdot \mathrm{Leb}$; witness measures are exactly the case $D = 1$. The defect governs the subject in four ways. (i) Every covering of K by planks parallel to the given directions satisfies $\sum \mathrm{rw} \geq 1/D_K(u)$ (Proposition 2.2). (ii) The classical bound $\sum \mathrm{rw} \geq 1/d$ is the statement that the *uniform* measure certifies $D_K \leq d$ for every direction family on every d-dimensional body (Theorem 2.5). (iii) D_K obeys a minimax (continuous linear-programming) duality (Theorem 7.7) whose feasible points are *certificates*: measures certify upper bounds, dual weight functions certify lower bounds — in particular the exact values in this paper come with matched primal–dual pairs. (iv) The sharp coverings of this paper are the configurations where $D = 1$ is attained and rigid, while the facet triple, where $D = \frac{3}{2}$ exactly (Theorem 7.5) but boundary tilings still give $\sum \mathrm{rw} \geq 1$ (Theorem 3.3), calibrates what transport alone cannot do.

Results and their hierarchy. *Main results (the triangle).* Theorem 6.11: for three pairwise non-parallel directions on Δ, a witness measure exists if and only if the three mid-level lines $\{u_i = \frac{1}{2}\}$ are concurrent — this settles the existence question of Gardner [5] *only* for three directions on the triangle. Theorems 6.3 and 6.6: for every concurrent cyclic triple, an explicit witness measure (weighted perimeter), an explicit tight covering attaining $\sum \mathrm{rw} = 1$, and its uniqueness — a two-parameter family of rigid sharp configurations extending the median case (Theorem 5.1).

Framework results (any convex body). The defect bound and its attainment (Proposition 2.2); the uniform-measure estimate $D_K \leq d$ (Theorem 2.5); the duality theorem with its explicit wedge certificates (§7); the exact facet defect $D_{\Delta^d}(\text{facets}) = \frac{d+1}{2}$ in every dimension (Theorem 8.2); the bound $\delta(u) \leq \frac{1}{6}$ on the concurrence defect of *any* direction family on the triangle (Theorem 7.3); and an explicit linear modulus for D near the concurrent locus (Theorem 7.15), which yields, for example, $\sum \mathrm{rw} \geq \frac{29}{31}$ for coverings by planks parallel to any cyclic triple with

$\max_i |\tau_i - \frac{1}{2}| \le \frac{1}{60}$ (Corollary 7.17); a numerical comparison with the general bound of [4] is deferred to the end of this introduction.

Illustrative cases. Two directions (Theorem 3.1), facet-parallel families (Theorem 3.3), and three facets plus one arbitrary plank (Theorem 4.1): calibration points where older methods are already sharp; they are not at the level of the main results.

Extensions and obstructions. A witness family on Δ^3 and the facet defect in higher dimensions (§8); the normalizer obstruction (Proposition 9.1, Remark 9.2, §9); and Appendix A, the classification of boundary tilings of cyclic triples (Proposition A.1) — the combinatorial engine behind both rigidity results.

A first genuinely tilted case. Beyond the concurrent locus, where the transport method no longer certifies the sharp constant, §10 frames the covering constants $C_\Delta \le C_\Delta^{111}$ as the remaining open objects. For one explicit triple — tilted, non-facet, non-concurrent — we establish $C_\Delta^{111} = 1$: every covering by three planks, one per direction, has $\sum \mathrm{rw} \ge 1$ (Theorem 10.8), in part by an exact-rational computation verified by an independent certificate checker. To our knowledge this is the first such genuinely tilted triple; it is a single triple (not the family), settles only the one-plank-per-direction constant (leaving C_Δ there open), and does not bear on Conjecture 1.1.

Comparison with the general bound. On the triangle the method of [4] gives $\sum \mathrm{rw} \ge 4\sqrt{3} - 6 \approx 0.928$ uniformly over *all* coverings by *all* plank families. Our stability bound $\sum \mathrm{rw} \ge \frac{29}{31} \approx 0.935$ (Corollary 7.17) exceeds this value, but only for coverings by planks parallel to the restricted direction families near the medians; Corollary 7.18 extends the comparison to an explicit neighbourhood of the whole concurrent family. This is a local consequence of the method, not an improvement of the general constant; see Remark 7.19 for the precise scope.

2. The transport defect and the uniform-measure bound

Definition 2.1. Let $K \subset \mathbb{R}^N$ be a convex body and $u_1, \dots, u_n$ directions on K (normalized affine coordinates $u_i : K \to [0,1]$, onto). The *transport defect* of the family is

$$D_K(u) \;=\; \inf\bigl\{D \ge 1 :\; \exists\, \mu \in \mathcal{P}(K) \text{ with } u_{i\#}\mu \le D \cdot \mathrm{Leb} \text{ on } [0,1] \text{ for all } i\bigr\}.$$

When $K = \Delta$ is the triangle we write $D(u)$. A measure feasible at level D will be called *D-admissible.*

Equivalently, in *filling* form: μ is D-admissible iff for each i there is a nonnegative measure ρ_i on $[0,1]$ (the *slack*) with $u_{i\#}\mu + \rho_i = D \cdot \mathrm{Leb}$; comparing total masses, ρ_i has mass $D - 1$. Thus $D_K(u) = 1$ iff the marginals can be made *exactly* uniform — iff a witness measure exists (the slack has mass 0).

Proposition 2.2. *(i) The infimum is attained. (ii) Every covering of K by finitely many planks, each parallel to one of the n directions, satisfies $\sum \mathrm{rw} \ge 1/D_K(u)$. (iii) $D_K(u) = 1$ if and only if a witness measure for $u_1, \dots, u_n$ exists. (iv) D_K is monotone: adding directions to the family cannot decrease it.*

Proof. (i) Take $D_m \downarrow D_K(u)$ and D_m-admissible μ_m; by weak-$*$ compactness of $\mathcal{P}(K)$ pass to a limit μ. For every open interval J the set $u_i^{-1}(J)$ is relatively open in K, so $\mu(u_i^{-1}(J)) \le \liminf_m \mu_m(u_i^{-1}(J)) \le D_K(u)|J|$; open intervals determine the bound $u_{i\#}\mu \le D_K(u)\,\mathrm{Leb}$ by outer regularity. (Taking $J = (-\delta, 1+\delta)$ also shows every feasible level is ≥ 1.) (ii) For a plank P parallel to u_i with normalized interval I and an attaining μ, $\mu(P \cap K) = u_{i\#}\mu(I) \le D_K(u)\,|I| \le D_K(u)\,\mathrm{rw}(P)$; summing over a covering gives $1 \le D_K(u) \sum \mathrm{rw}$. (iii) If $D_K(u) = 1$, an attaining μ has $u_{i\#}\mu \le \mathrm{Leb}$ with both sides of total mass 1, hence $u_{i\#}\mu = \mathrm{Leb}$: a witness. The converse is the case $D = 1$ of the definition. (iv) is immediate: admissibility for the larger family implies it for the smaller. ∎

The quantity the conjecture is actually about deserves its own name.

Definition 2.3. The *covering constant* of a direction family u on K is

$$C_K(u) \;=\; \inf\Big\{\textstyle\sum_i \mathrm{rw}_K(P_i):\; K\subset\bigcup_i P_i,\ \text{each } P_i \text{ a plank parallel to some } u_j\Big\},$$

with *any* number of planks. When $u=(u_1,u_2,u_3)$ is a triple we also write $C_K^{111}(u)$ for the restricted infimum in which *exactly one* plank is taken per direction:

$$C_K^{111}(u) \;=\; \inf\Big\{\,|I_1|+|I_2|+|I_3|:\; K\subset\bigcup_{i=1}^{3}\{u_i\in I_i\}\Big\}.$$

Remark 2.4. The three constants are ordered:

$$\frac{1}{D_K(u)} \;\le\; C_K(u) \;\le\; C_K^{111}(u) \;\le\; 1.$$

The first inequality is Proposition 2.2(ii); the second holds because a one-plank-per-direction covering is in particular a covering (fewer admissible coverings, larger infimum); the last is the single plank $\{u_1\in[0,1]\}=K$. Conjecture 1.1, restricted to planks in the directions of u, is exactly $C_K(u)=1$; the stronger $C_K^{111}(u)=1$ is the three-plank version. Two *gaps* organize the program: $G_{\mathrm{tr}}(u)=C_K(u)-1/D_K(u)$ (how far transport alone is from the covering truth) and $G_{\mathrm{mult}}(u)=C_K^{111}(u)-C_K(u)$ (what extra planks buy). On the concurrent families of §§5–6 all three constants equal 1 and both gaps vanish (the tight covering already uses one plank per direction), while at the facet triple $1/D=\frac{2}{3}$, $C_\Delta=1$ (tiling, Theorem 3.3), so $G_{\mathrm{tr}}=\frac{1}{3}$. Section 10 collects what is known and formulates the program; Theorem 10.8 determines C_Δ^{111} at one non-concurrent triple, leaving C_Δ there open.

The classical bound $\sum\mathrm{rw}\ge 1/d$ is, in this language, a statement about one particular certificate: the uniform measure.

Theorem 2.5. *Let $d\ge 2$ and let $K\subset\mathbb{R}^d$ be a convex body. For* every *finite family of directions u on K, the uniform probability measure on K is d-admissible; hence $D_K(u)\le d$, and every covering of K by planks satisfies $\sum_i \mathrm{rw}_K(P_i)\ge 1/d$. The per-plank estimate $\mu(P)\le d\,\mathrm{rw}_K(P)$ for the uniform measure μ is sharp at the simplex; hence the argument, run with the uniform measure, cannot improve the constant $1/d$. (For $d=1$ the conjecture itself is trivial: intervals covering an interval satisfy $\sum\mathrm{rw}\ge 1$.)*

Proof. Let μ be the uniform probability measure on K. Fix a direction, i.e. a linear functional normalized so that $\langle u,\cdot\rangle\in[0,w]$ on K, $w=\mathrm{w}_K(u)$. The marginal density is $f(t)=V(t)/\operatorname{vol}(K)$ with $V(t)=\operatorname{vol}_{d-1}(K\cap\{\langle u,\cdot\rangle=t\})$; by Brunn–Minkowski $g:=V^{1/(d-1)}$ is concave on $[0,w]$. If g peaks at t^* with value g^*, then $g\ge g^*h$ for the tent h ($h(0)=h(w)=0$, $h(t^*)=1$), so $f=g^{d-1}\ge(\max f)\,h^{d-1}$ and $1=\int f\ge(\max f)\int_0^w h^{d-1}=(\max f)\,w/d$, giving $\max f\le d/w$. In normalized coordinates this says exactly $u_\#\mu\le d\cdot\mathrm{Leb}$ for every direction u at once, so μ is d-admissible for every family and $D_K(u)\le d$; the covering bound is Proposition 2.2(ii). For sharpness of the per-plank estimate: equality in $\max f\le d/w$ forces $V\propto h^{d-1}$ (conical sections), realized by the simplex $K=\Delta^d$ with u a vertex–facet direction, where a thin plank at the facet has $\mu(P)=(d-o(1))\,\mathrm{rw}_K(P)$. Thus for the simplex the *uniform* measure cannot certify a constant larger than $1/d$ by the union bound; this does *not* assert any covering of the simplex with $\sum\mathrm{rw}$ near $1/d$ (indeed Theorem 3.3 gives $\sum\mathrm{rw}\ge 1$ for facet-parallel coverings). ☐

Remark 2.6. Other measures do better for *restricted* direction families: the witness measures of §6 certify the full constant 1 for suitable triples, and §7 quantifies the best constant certifiable per triple. For *all* directions simultaneously, however, no single measure certifies the constant

1: uniform marginals in all directions would force $\mathbb{E}[x_k] = \frac{1}{2}$ against $\sum x_k = 1$ (Gardner's obstruction [5]). On the triangle the best constant certifiable by one measure against all directions at once lies between $\frac{1}{2}$ (the uniform measure certifies $D \le 2$ for all directions simultaneously, Theorem 2.5 with $d = 2$) and $\frac{2}{3}$ (the facet triple already forces it; Theorem 7.5); we do not pursue its exact value.

3. Sharp cases: two directions and facet-parallel families

Theorem 3.1. *If every plank covering the triangle is parallel to one of two fixed* non-parallel *directions, then* $\sum \mathrm{rw} \ge 1$*, and this is sharp. (If the two directions are parallel the statement is trivial: all planks share one normalized coordinate* f*, the intervals* $f(P_i)$ *cover* $[0,1]$*, and* $\sum \mathrm{rw} = \sum |I_i| \ge 1$*.)*

Proof. We use Gardner's existence theorem [5, Theorem 1], whose exact statement is: *for a convex body* $K \subset \mathbb{R}^n$ *and two directions* θ_1, θ_2*, there exists a relative-width measure on* K *for* $\{\theta_1, \theta_2\}$ — a probability measure ν, possibly singular, such that $\nu(S \cap K)$ equals the relative width of S for every slab S orthogonal to θ_1 or θ_2; equivalently, the pushforward of ν under each normalized affine coordinate $f_{\theta_i} : K \to [0,1]$ is Leb$[0,1]$. Applying this to $K = \Delta$ and our two directions gives ν with $f_{u\#}\nu = f_{v\#}\nu = \mathrm{Leb}$. Then $\nu(P) = |I| = \mathrm{rw}(P)$ for each plank, and $1 = \nu(\Delta) \le \sum \nu(P_i) = \sum \mathrm{rw}(P_i)$. □

The facet-parallel case rests on a one-dimensional sumset estimate, which we isolate.

Lemma 3.2 (sumset lemma)**.** *Let* $C_1, \dots, C_{d+1}$ *be nonempty compact subsets of* $[0,1]$ *with* $\sum_k |C_k| > d$*. Then* $1 \in C_1 + \cdots + C_{d+1}$ *(Minkowski sum).*

Proof. The one-dimensional Brunn–Minkowski inequality $|A+B| \ge |A|+|B|$ holds for nonempty compact $A, B \subset \mathbb{R}$: $A + B \supseteq (A + \min B) \cup (\max A + B)$, and the two translates meet only at $\max A + \min B$. Iterating it, $E := C_1 + \cdots + C_d$ (compact) satisfies $|E| \ge \sum_{k=1}^{d} |C_k|$; as $E \subseteq [0,d]$, $|[0,1] \setminus E| \le |[0,d] \setminus E| = d - |E|$, whence $|E \cap [0,1]| \ge 1 - d + \sum_{k=1}^{d} |C_k|$. Also $1 - C_{d+1} \subseteq [0,1]$ has measure $|C_{d+1}|$. Both sets lie in $[0,1]$ and their measures sum to $1 + \left(\sum_{k=1}^{d+1} |C_k| - d\right) > 1$, so they intersect: there is $t \in E$ with $1 - t \in C_{d+1}$, i.e. $1 \in C_1 + \cdots + C_{d+1}$. □

Theorem 3.3. *Planks parallel to the facets of a simplex* Δ^d*, in any number, that cover* Δ^d *satisfy* $\sum \mathrm{rw} \ge 1$*, and this is sharp.*

Proof. Write $\Delta^d = \{\lambda \in [0,1]^{d+1} : \sum_k \lambda_k = 1\}$ in barycentric coordinates; a plank parallel to facet k is a coordinate slab $\{\lambda_k \in I\}$. Grouping the planks by facet, let $U_k = \bigcup_{c(i)=k} I_i \subset [0,1]$ be the forbidden set on facet k, so $\sum_k |U_k| \le \sum_i |I_i| =: R$, and set $C_k = [0,1] \setminus U_k$. A point $\lambda \in \Delta^d$ is uncovered iff $\lambda_k \in C_k$ for every k; since $C_k \subset [0,1]$, an uncovered point exists *iff* $1 \in C_1 + \cdots + C_{d+1}$.

Assume $R < 1$; we produce an uncovered point. Each U_k is a finite union of closed intervals, so C_k is a finite union of intervals, relatively open in $[0,1]$, with $|C_k| = 1 - |U_k| \ge 1 - R > 0$; it need not be compact, so fix $\eta = \frac{1-R}{d+2} > 0$ and choose a *compact* $C'_k \subseteq C_k$ (shave each component) with $|C'_k| \ge |C_k| - \eta$. Then

$$\textstyle\sum_{k=1}^{d+1} |C'_k| \;\ge\; (d+1) - R - (d+1)\eta \;=\; d + (1-R) - (d+1)\eta \;>\; d,$$

since $(d+1)\eta < 1 - R$. Lemma 3.2 applies: $1 \in C'_1 + \cdots + C'_{d+1}$, i.e. $1 = \lambda_1 + \cdots + \lambda_{d+1}$ with $\lambda_k \in C'_k \subseteq C_k$ — an uncovered $\lambda \in \Delta^d$. Contrapositively, covering forces $R \ge 1$. Sharpness: the facet tiling $I_k = [0, \frac{1}{d+1}]$ gives $\sum \mathrm{rw} = 1$. □

Remark 3.4. The edge-by-edge argument does *not* prove this: on an edge only two barycentric coordinates vary, so no single edge can force $\sum_k |U_k| \geq 1$ across all $d+1$ facets. The Minkowski-sum reduction above is the correct route; note that no witness *measure* can prove it either, since facet directions violate the concurrence condition of Proposition 6.1.

4. Three facets plus one arbitrary plank

This section records an illustrative mixed case — a fibration argument that handles one arbitrary direction on top of the facet triple. It is a calibration point, not at the level of the main results of §§5–6.

Theorem 4.1. *Let $P_i = \{\lambda_i \in I_i\}$ $(i = 1, 2, 3)$ be facet-parallel with $a_i = |I_i|$, and let $Q = \{f \in J\}$, $t = |J|$, with $f : \Delta \to [0, 1]$ affine normalized. If $\Delta \subset P_1 \cup P_2 \cup P_3 \cup Q$ then $a_1 + a_2 + a_3 + t \geq 1$.*

Proof. Choose an active edge of f; relabelling, $f(v_j) = 0$, $f(v_k) = 1$, and let v_i be the opposite vertex, $\theta = f(v_i)$. For $s \in [0, 1]$ consider the fibre $F_s = \{\lambda_i = s\}$, parametrized by $u = \lambda_k \in [0, 1 - s]$, so $\lambda_j = 1 - s - u$ and $f|_{F_s}(u) = \theta s + u$. On F_s: $P_i \cap F_s = F_s$ if $s \in I_i$ (else $\varnothing$); $P_j \cap F_s$, $P_k \cap F_s$, $Q \cap F_s$ have lengths $\leq a_j, a_k, t$. Put $B = a_j + a_k + t$. If $s \notin I_i$ and F_s is covered, its length $1 - s \leq B$. If $B \geq 1$ we are done; else $[0, 1 - B) \subset I_i$, so $a_i \geq 1 - B$. In both cases $a_1 + a_2 + a_3 + t = a_i + B \geq 1$. □

5. The median theorem and its rigidity

Let m_1, m_2, m_3 be the medians of Δ as normalized affine forms: with A, B, C the vertices,

$$m_1 = (\tfrac{1}{2}, 0, 1), \quad m_2 = (0, 1, \tfrac{1}{2}), \quad m_3 = (1, \tfrac{1}{2}, 0),$$

(vertex values). One has $m_1 + m_2 + m_3 \equiv \frac{3}{2}$. A *median plank* is $P_i = \{m_i \in I_i\}$, $I_i = [l_i, h_i]$, $r_i = |I_i| > 0$.

Theorem 5.1. *If three median planks cover Δ, then $r_1 + r_2 + r_3 \geq 1$, with equality if and only if $I_1 = I_2 = I_3 = [\frac{1}{3}, \frac{2}{3}]$.*

We are not aware of a previous tight *non-facet* three-direction rigidity result for the triangle: the sharp cases recorded in the survey [7] (and in [5, 4]) concern facet/coordinate directions, two directions, or symmetric bodies.

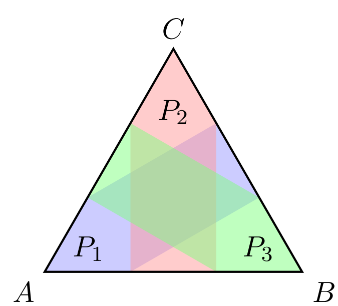

Figure 1. The tight median covering of Theorem 5.1: $I_1 = I_2 = I_3 = [\frac{1}{3}, \frac{2}{3}]$. Each plank $P_i = \{m_i \in [\frac{1}{3}, \frac{2}{3}]\}$ is the central-thirds band of its median; the three traces tile each edge, and the planks cover Δ with $\sum \mathrm{rw} = 1$.

Proof of the bound. Let μ be the perimeter measure: uniform in the parameter on each of the three edges, mass $\frac{1}{3}$ each. A direct computation shows each pushforward $m_{i\#}\mu = \mathrm{Leb}[0,1]$: μ is a witness measure, certifying $D(m_1, m_2, m_3) = 1$ in the language of Definition 2.1. Indeed for m_1: on AB, m_1 ranges over $[0, \frac{1}{2}]$ with Jacobian 2 (density $\frac{1}{3} \cdot 2$); on CA, over $[\frac{1}{2}, 1]$ likewise; on BC, over $[0,1]$ with Jacobian 1 (density $\frac{1}{3}$); the total density is 1 on $[0,1]$. Hence $\mu(P_i) = \mathrm{Leb}(I_i) = r_i$, and $1 = \mu(\Delta) \le \sum \mu(P_i) \le \sum r_i$. ☐

Proof of rigidity. Equality in $1 \le \sum \mu(P_i)$ forces the P_i to partition μ a.e.; since μ lives on the edges, *on each edge the three traces tile* $[0,1]$. We claim there are exactly three such edge-tilings, and only one covers the interior.

Three edge-tilings. On each edge each median is affine, so the three traces are explicit intervals depending piecewise-linearly on (l_i, h_i) through $a_i = |I_i \cap [0, \frac{1}{2}]|$ and $b_i = |I_i \cap [\frac{1}{2}, 1]|$. Classifying each I_i by crossing type (contained in $[0, \frac{1}{2}]$, crossing $\frac{1}{2}$, or contained in $[\frac{1}{2}, 1]$) and each edge by the left-to-right order of its traces, the abutting (tiling) conditions become a finite family of linear systems in (l_i, h_i).

Lemma 5.2. *The edge-tiling conditions have exactly three solutions with $0 \le l_i < h_i \le 1$: $[\frac{1}{3}, \frac{2}{3}]^3$, $[0, \frac{1}{3}]^3$, and $[\frac{2}{3}, 1]^3$.*

This is the case $\tau_i = \frac{1}{2}$ of Proposition A.1, proved in full in Appendix A for arbitrary cyclic triples: the symmetries of the configuration reduce the a priori 27 crossing patterns to seven classes, five of which are impossible and two of which lead to cyclic linear systems with unique solutions. The lemma has also been verified by three independent exact computer enumerations, archived as ancillary files.

Interior witness. The centroid G has $m(G) = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Since $\frac{1}{2} \notin [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$, the tilings $[0, \frac{1}{3}]^3$ and $[\frac{2}{3}, 1]^3$ leave G uncovered; they tile the boundary but are not coverings of Δ. The tiling $[\frac{1}{3}, \frac{2}{3}]^3$ covers Δ and has $\sum r_i = 1$. Hence the unique covering with $\sum r_i = 1$ is $[\frac{1}{3}, \frac{2}{3}]^3$. ☐

6. A dimension-agnostic necessary condition for transport

The proofs of Theorems 3.1 and 5.1 share one device: a probability measure whose pushforward under each direction is uniform. Which direction-tuples admit one?

Proposition 6.1. *Let $u_1, \dots, u_{d+1}$ be normalized affine forms on Δ^d with vertex-value matrix V invertible. If a probability measure μ on Δ^d has $u_{i\#}\mu = \mathrm{Leb}[0,1]$ for all i, then its barycenter $p \in \Delta^d$ satisfies $Vp = \frac{1}{2}\mathbf{1}$; equivalently the mid-level hyperplanes $\{u_i = \frac{1}{2}\}$ concur at p, which requires $\mathbf{1}^\top V^{-1}\mathbf{1} = 2$ and $V^{-1}\mathbf{1} \ge 0$.*

Proof. $\frac{1}{2} = \mathbb{E}_\mu[u_i] = \sum_j V_{ij}\mathbb{E}_\mu[x_j] = (Vp)_i$, and $p \in \Delta^d$ gives $\mathbf{1}^\top p = 1$, $p \ge 0$; substitute $p = \frac{1}{2}V^{-1}\mathbf{1}$. ☐

The three medians satisfy this with p the centroid; the three facets do not ($\mathbf{1}^\top V^{-1}\mathbf{1} = 3 \ne 2$), recovering Gardner's obstruction. The concurrence family is two-dimensional (parametrized by p): a normalized direction has vertex values $\{0, \tau, 1\}$, and $Vp = \frac{1}{2}\mathbf{1}$ ties each τ_i to p. The medians are the case $p =$ centroid, forcing $\tau_i = \frac{1}{2}$. Among these, the explicit perimeter measure singles out the medians:

Proposition 6.2. *For a direction u with vertex values $\{0, \tau, 1\}$, the perimeter measure μ (uniform on each edge, mass $\frac{1}{3}$) satisfies $u_{\#}\mu = \mathrm{Leb}[0,1]$ if and only if $\tau = \frac{1}{2}$.*

Proof. On the three edges u is affine between the pairs $\{0,\tau\}, \{\tau, 1\}, \{0, 1\}$, so $u_{\#}\mu$ has density $\frac{1}{3\tau} + \frac{1}{3}$ on $[0, \tau]$ and $\frac{1}{3(1-\tau)} + \frac{1}{3}$ on $[\tau, 1]$. Both equal 1 iff $\tau = \frac{1}{2}$. ☐

Uniform edge-weights thus single out the medians. Allowing *unequal* edge masses, however, the construction covers a full two-parameter family:

Theorem 6.3 (weighted perimeter)**.** *Let $p = (p_A, p_B, p_C)$ be interior with $\max_j p_j < \frac{1}{2}$, and let $u_1 = (\tau_1, 0, 1)$, $u_2 = (0, 1, \tau_2)$, $u_3 = (1, \tau_3, 0)$ (vertex values at A, B, C) be the cyclic directions concurrent at p, i.e. $\tau_1 = \frac{1/2 - p_C}{p_A}$, $\tau_2 = \frac{1/2 - p_B}{p_C}$, $\tau_3 = \frac{1/2 - p_A}{p_B}$. Let μ_p be uniform on each edge with masses $w_{AB} = 1 - 2p_C$, $w_{BC} = 1 - 2p_A$, $w_{CA} = 1 - 2p_B$ (summing to 1). We fix once and for all the* mass convention

$$\alpha := w_{BC} = 1 - 2p_A, \qquad \beta := w_{CA} = 1 - 2p_B, \qquad \gamma := w_{AB} = 1 - 2p_C :$$

each Greek letter names the mass of the edge opposite *the like-named vertex. This convention is used verbatim in Theorem 6.6 and in §7. Then $u_{i\#}\mu_p = \mathrm{Leb}[0, 1]$ for $i = 1, 2, 3$; consequently every covering of Δ by planks in these three directions satisfies $\sum \mathrm{rw} \geq 1$.*

Proof. Each u_i is affine of full range $[0, 1]$ on the edge joining its 0- and 1-vertices (u_1 on BC, u_2 on AB, u_3 on CA) and affine of range $[0, \tau_i]$, resp. $[\tau_i, 1]$, on the other two. The pushforward of a uniform edge mass w_e through an affine map of range length L is uniform with density w_e/L. Hence $u_{1\#}\mu_p$ has density $w_{BC} + w_{AB}/\tau_1$ on $[0, \tau_1]$ and $w_{BC} + w_{CA}/(1 - \tau_1)$ on $[\tau_1, 1]$, and similarly for u_2, u_3: six equations. Using $p_A + p_B + p_C = 1$, each composite term collapses by the identity $(1 - 2x)\,p/(\frac{1}{2} - x) = 2p$: e.g. $w_{AB}/\tau_1 = (1 - 2p_C)p_A/(\frac{1}{2} - p_C) = 2p_A$, so $w_{BC} + w_{AB}/\tau_1 = (1 - 2p_A) + 2p_A = 1$; likewise $1 - \tau_1 = \frac{1/2 - p_B}{p_A}$ gives $w_{CA}/(1 - \tau_1) = 2p_A$. All six densities equal 1, so each marginal is $\mathrm{Leb}[0, 1]$, and the transport argument of Theorem 5.1 applies verbatim. Finally $\tau_i \in (0, 1)$ for all i iff $\max_j p_j < \frac{1}{2}$ iff all three masses are positive, so every valid cyclic triple admits μ_p. □

Taking p the centroid recovers Theorem 5.1's bound ($\tau_i = \frac{1}{2}$, equal masses). For p off the centroid the directions are tilted non-median triples; e.g. $p = (\frac{9}{20}, \frac{3}{10}, \frac{1}{4})$ gives $\tau = (\frac{5}{9}, \frac{4}{5}, \frac{1}{6})$ with masses $(\frac{1}{2}, \frac{1}{10}, \frac{2}{5})$. Theorem 6.3 thus proves the bound of Conjecture 1.1 for a two-parameter family of direction triples on the triangle, answering the finite-direction existence question of [5] affirmatively for the cyclic concurrent family. Figure 2 shows the running example.

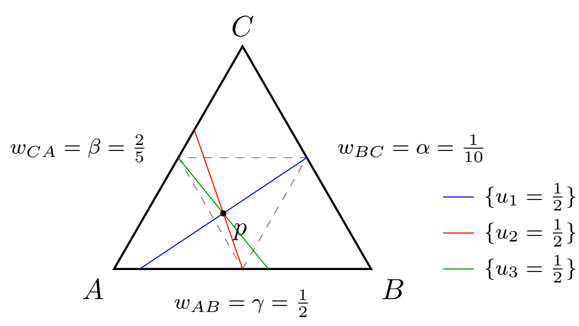

FIGURE 2. The weighted-perimeter witness of Theorem 6.3 at $p = (\frac{9}{20}, \frac{3}{10}, \frac{1}{4})$, i.e. $\tau = (\frac{5}{9}, \frac{4}{5}, \frac{1}{6})$: the three mid-level lines $\{u_i = \frac{1}{2}\}$ concur at p, which lies in the open medial triangle (dashed), and μ_p puts the indicated masses uniformly on the edges.

Combining necessity (Proposition 6.1) with the explicit construction (Theorem 6.3) yields a *characterization*:

Corollary 6.4. *Call a direction triple* cyclic *if its vertex values are $u_1 = (\tau_1, 0, 1)$, $u_2 = (0, 1, \tau_2)$, $u_3 = (1, \tau_3, 0)$ with $\tau_i \in (0, 1)$. A cyclic triple admits a uniform-marginal witness measure if and only if the three mid-level lines $\{u_i = \frac{1}{2}\}$ concur. In that case the concurrence point p automatically lies in the open medial triangle ($p_j > 0$ and $p_j < \frac{1}{2}$ for all j), and the weighted-perimeter measure μ_p of Theorem 6.3 is a witness.*

Proof. The vertex-value matrix has $\det V = -(1 + \tau_1\tau_2\tau_3) \neq 0$, so Proposition 6.1 applies and gives necessity. Conversely, if the lines concur at p (a point of the affine plane of Δ, so $\sum_j p_j = 1$), then p is the unique solution of $Vp = \frac{1}{2}\mathbf{1}$, which in closed form reads $2(1 + \tau_1\tau_2\tau_3)\, p_A = 1 - \tau_3(1 - \tau_2)$ and cyclically; each right-hand side is positive for $\tau_i \in (0, 1)$, so $p_j > 0$, and the identities $\frac{1}{2} - p_C = \tau_1 p_A$, $\frac{1}{2} - p_A = \tau_3 p_B$, $\frac{1}{2} - p_B = \tau_2 p_C$ give $p_j < \frac{1}{2}$. Theorem 6.3 then supplies μ_p. □

Remark 6.5. The *anti-cyclic* pattern (all 0- and 1-roles exchanged) reduces to the cyclic one by a reflection of Δ (a transposition of two vertices), which maps the medial triangle to itself; Theorem 6.3, Corollary 6.4 and Theorem 6.6 below hold for it verbatim. All remaining role assignments are classified by Theorem 6.11 below.

The bound of Theorem 6.3 is in fact *attained* at every member of the family, and the tight covering is unique — a two-parameter family of sharp configurations extending the median rigidity of §5.

The coverage argument below rests on an elementary topological fact, which we isolate.

Pocket Lemma. Let H be an open convex subset of the plane of Δ such that $P := H \cap \operatorname{int} \Delta \neq \varnothing$ and the *closure* $\overline{P}$ of P meets $\partial\Delta$ in a finite set. Then $H \subseteq \overline{\Delta}$; in particular H is bounded.

Proof. Suppose $y \in H \setminus \overline{\Delta}$, and choose a disk $B \subseteq P$. For $z \in B$ the segment $[z, y]$ lies in H by convexity and joins $\operatorname{int}\Delta$ to the complement of $\overline{\Delta}$; let $w(z)$ be the first point of $\partial\Delta$ on it, starting from z. Then $[z, w(z)) \subseteq H \cap \operatorname{int}\Delta$, so $w(z) \in \overline{P} \cap \partial\Delta$, a finite set. But for a fixed boundary point w, every z with $w(z) = w$ lies on the line through y and w, and finitely many lines cannot cover the disk B — a contradiction. □

Theorem 6.6 (tightness and rigidity of the cyclic family)**.** *Let p and u_1, u_2, u_3 be as in Theorem 6.3, with the mass convention fixed there: $\alpha = w_{BC} = 1 - 2p_A$, $\beta = w_{CA} = 1 - 2p_B$, $\gamma = w_{AB} = 1 - 2p_C$ ($\alpha, \beta, \gamma > 0$, $\alpha + \beta + \gamma = 1$). Set $S = \alpha\beta + \beta\gamma + \gamma\alpha$. Then the planks $P_i = \{u_i \in I_i\}$ with*

$$I_1 = \left[\tfrac{\alpha\gamma}{S},\, 1 - \tfrac{\alpha\beta}{S}\right], \qquad I_2 = \left[\tfrac{\beta\gamma}{S},\, 1 - \tfrac{\alpha\gamma}{S}\right], \qquad I_3 = \left[\tfrac{\alpha\beta}{S},\, 1 - \tfrac{\beta\gamma}{S}\right]$$

cover Δ, with $\sum \mathrm{rw} = \frac{\beta\gamma}{S} + \frac{\alpha\beta}{S} + \frac{\alpha\gamma}{S} = 1$. Moreover this is the unique covering of Δ with $\sum \mathrm{rw} = 1$ by three planks of positive width, one in each of the directions u_1, u_2, u_3. At p the centroid ($\alpha = \beta = \gamma = \frac{1}{3}$) it is $[\frac{1}{3}, \frac{2}{3}]^3$, recovering Theorem 5.1.

The proof rests on three lemmas, stated with the notation of Theorem 6.6. Throughout, $l = (l_1, l_2, l_3) = (\alpha\gamma, \beta\gamma, \alpha\beta)/S$ and $h = (h_1, h_2, h_3) = \mathbf{1} - (\alpha\beta, \alpha\gamma, \beta\gamma)/S$ denote the interval endpoints, so $I_i = [l_i, h_i]$, and $\tau_1 = \frac{\gamma}{1-\alpha}$, $\tau_2 = \frac{\beta}{1-\gamma}$, $\tau_3 = \frac{\alpha}{1-\beta}$ from the formulas of Theorem 6.3. A direct substitution shows $(l, h) = (\lambda, \rho)$, the first solution of Proposition A.1; in particular $l_i < \tau_i < h_i$, and the traces of P_1, P_2, P_3 tile all three edges, so $\partial\Delta \subset \bigcup P_i$.

Lemma 6.7 (margin identity)**.** *Let $a_1 = \alpha(1 - \alpha)$, $a_2 = \gamma(1 - \gamma)$, $a_3 = \beta(1 - \beta)$ and $\Lambda = a_1u_1 + a_2u_2 + a_3u_3$. Then $\Lambda \equiv S$ on the plane of Δ, $\sum_i a_i = 2S$, and*

$$\textstyle\sum_i a_i l_i = S - \frac{\alpha\beta\gamma}{S} < S < S + \frac{\alpha\beta\gamma}{S} = \sum_i a_i h_i. \tag{1}$$

Proof. At the vertices: $\Lambda(A) = a_1\tau_1 + a_3 = \alpha\gamma + \beta(\alpha+\gamma) = S$, $\Lambda(B) = a_2 + a_3\tau_3 = \gamma(\alpha+\beta) + \alpha\beta = S$, $\Lambda(C) = a_1 + a_2\tau_2 = \alpha(\beta+\gamma) + \beta\gamma = S$. An affine function agreeing at three affinely independent points is constant: $\Lambda \equiv S$ on the plane of Δ. Moreover $\sum_i a_i = 1 - (\alpha^2 + \beta^2 + \gamma^2) = 2S$, and expanding (using $1 - \alpha = \beta + \gamma$ etc.)

$$\sum_i a_i l_i = \frac{\alpha^2\gamma(\beta+\gamma) + \beta\gamma^2(\alpha+\beta) + \alpha\beta^2(\gamma+\alpha)}{S} = \frac{(\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2) + \alpha\beta\gamma}{S} = \frac{S^2 - \alpha\beta\gamma}{S},$$

since $S^2 = \sum \alpha^2\beta^2 + 2\alpha\beta\gamma$; the same expansion gives $\sum_i a_i(1 - h_i) = \frac{S^2 - \alpha\beta\gamma}{S}$, whence $\sum_i a_i h_i = 2S - \frac{S^2 - \alpha\beta\gamma}{S} = S + \frac{\alpha\beta\gamma}{S}$. □

Lemma 6.8 (coverage)**.** *The planks $P_i = \{u_i \in [l_i, h_i]\}$, $i = 1, 2, 3$, cover Δ.*

Proof. Suppose not; the uncovered set U is relatively open (planks are closed) and misses the tiled boundary $\partial\Delta$, and it decomposes as $U = \bigcup_\sigma P_\sigma$ over the eight sign patterns $\sigma \in \{\mathrm{lo}, \mathrm{hi}\}^3$, where $P_\sigma = H_\sigma \cap \operatorname{int}\Delta$ and H_σ is the open convex set $\{u_i < l_i\ (\sigma_i = \mathrm{lo}),\ u_i > h_i\ (\sigma_i = \mathrm{hi})\}$; write c_i for the active bound (l_i or h_i). We show each P_σ is empty.

First, $\overline{P_\sigma} \cap \partial\Delta$ is finite. Indeed, such a point x satisfies the closed pattern constraints ($u_j(x) \le l_j$ if $\sigma_j = \mathrm{lo}$, $u_j(x) \ge h_j$ if $\sigma_j = \mathrm{hi}$), and, lying on the tiled boundary, is covered by some P_j, so $u_j(x) \in [l_j, h_j]$; together these force $u_j(x) = c_j$ for that j. Each u_j is affine and nonconstant on each edge (its two vertex values there are distinct), so $\{u_j = c_j\}$ meets each edge in at most one point: at most $3 \times 3 = 9$ points in all — this count includes the vertices of Δ (each lies on two edges) and only drops if several of these constraint points coincide. By the Pocket Lemma applied to H_σ (whose intersection with $\operatorname{int}\Delta$ is P_σ), $H_\sigma \subseteq \overline{\Delta}$ whenever $P_\sigma \ne \varnothing$; in particular H_σ is bounded.

No two of the lines $\ell_i = \{u_i = c_i\}$ are parallel: $u_i \parallel u_j$ would force $\tau_1(1-\tau_2) = 1$, $\tau_2(1-\tau_3) = 1$ or $\tau_3(1-\tau_1) = 1$, impossible for $\tau \in (0,1)^3$. If the three lines are concurrent, H_σ is an intersection of three open half-planes whose boundaries pass through one point, hence empty or an unbounded open cone; boundedness gives $H_\sigma = \varnothing$. Otherwise H_σ, if nonempty, is the open triangle with vertices $v_{ij} = \ell_i \cap \ell_j \in \overline{H_\sigma} \subseteq \overline{\Delta}$, and each v_{ij} satisfies the closed constraint in the third index k. Since $\Lambda \equiv S$ (Lemma 6.7), $u_k(v_{ij}) = (S - a_i c_i - a_j c_j)/a_k$, so that constraint reads $S \le \sum_m a_m c_m$ if $\sigma_k = \mathrm{lo}$, and $S \ge \sum_m a_m c_m$ if $\sigma_k = \mathrm{hi}$. For a *mixed* pattern both directions occur, forcing $S = \sum_m a_m c_m$; but then $u_k(v_{ij}) = c_k$, i.e. the three lines concur — excluded. For $\sigma = (\mathrm{lo}, \mathrm{lo}, \mathrm{lo})$ we get $S \le \sum a_i l_i$, and for $\sigma = (\mathrm{hi}, \mathrm{hi}, \mathrm{hi})$, $S \ge \sum a_i h_i$ — both contradict (1). Hence $U = \varnothing$. □

Lemma 6.9 (interior witnesses)**.** *Neither $(\,[0, l_i]\,)_i$ nor $(\,[h_i, 1]\,)_i$ covers Δ: with $\delta = \frac{\alpha\beta\gamma}{2S^2}$, the point x^* with $u_i(x^*) = l_i + \delta$ for all i and the point y^* with $u_i(y^*) = h_i - \delta$ for all i are interior points of Δ uncovered by the respective triple.*

Proof. The prescription $u_i(x^*) = l_i + \delta$ is consistent with $\Lambda \equiv S$ by (1) and $\sum a_i = 2S$, and determines a unique point of the plane since the vertex-value matrix V is invertible. Its barycentric coordinates are

$$x^* = \frac{1}{2S^2}\Big(\alpha\beta(1-\alpha)(1+\alpha-\beta),\ \beta\gamma(1-\beta)(\alpha+2\beta),\ \alpha\gamma(\alpha+\beta)(2-2\alpha-\beta)\Big),$$

all positive: $\alpha, \beta, \gamma > 0$ handle the leading factors (with $1 - \alpha = \beta + \gamma > 0$), while $1 + \alpha - \beta > 1 - \beta > 0$, $\alpha + 2\beta > 0$, and $2 - 2\alpha - \beta = \beta + 2\gamma > 0$ using $\alpha + \beta + \gamma = 1$. So $x^* \in \operatorname{int}\Delta$ and $u_i(x^*) > l_i$ for every i: x^* is uncovered by $(\,[0, l_i]\,)$. Symmetrically the point y^* with $u_i(y^*) = h_i - \delta$, with coordinates $\frac{1}{2S^2}\big(\alpha\gamma(1-\alpha)(2\alpha+\beta),\ \alpha\beta(1-\beta)(1+\beta-\alpha),\ \beta\gamma(\alpha+\beta)(2-\alpha-2\beta)\big)$, positive likewise ($2\alpha + \beta > 0$, $1 + \beta - \alpha > 1 - \alpha > 0$, $2 - \alpha - 2\beta = \alpha + 2\gamma > 0$), is uncovered by $(\,[h_i, 1]\,)$. □

Proof of Theorem 6.6. Coverage and value. Lemma 6.8 gives the covering; $\sum \mathrm{rw} = \sum(h_i - l_i) = 1$ is the displayed computation in the statement.

Uniqueness. Let (I'_i) be any covering with all $r'_i = |I'_i| > 0$ and $\sum r'_i = 1$. Since $u_{i\#}\mu_p = \mathrm{Leb}[0,1]$ (Theorem 6.3), $\mu_p(P'_i) = r'_i$, and $1 = \mu_p(\Delta) \le \sum \mu_p(P'_i) = 1$ forces the planks to partition μ_p a.e.; as μ_p has positive uniform density on each edge, the traces tile all three edges. By Proposition A.1, (I'_i) is $(\,[\lambda_i,\rho_i]\,) = (I_i)$, $(\,[0,\lambda_i]\,)$ or $(\,[\rho_i,1]\,)$. The last two do not cover Δ by Lemma 6.9. Hence $(I'_i) = (I_i)$. ∎

Remark 6.10. At the centroid, $\delta = \frac{1}{6}$ and both witnesses degenerate to the point with $u_i \equiv \frac{1}{2}$ — the centroid itself, recovering the interior witness used in the proof of Theorem 5.1. The identities of Lemma 6.7 and the positivity of the witness coordinates in Lemma 6.9, as well as exact coverage checks at several family points, are verified symbolically in the archived ancillary script `cyclic_family_tight_rigidity.py`.

Finally, the restriction to cyclic role assignments can be removed altogether. The plank problem is invariant under the *flip* of a normalized direction, $u \mapsto 1-u$, $I \mapsto 1-I$ (the planks, their widths, and uniformity of marginals are unchanged, and each mid-level line is fixed). Call the edge joining the 0- and 1-vertices of a direction its *full edge*. Flips reduce every role assignment to one of three shapes, and the existence question is settled in each:

Theorem 6.11 (three-direction characterization)**.** *Let u_1, u_2, u_3 be normalized directions on Δ, pairwise non-parallel. A probability measure μ on Δ with $u_{i\#}\mu = \mathrm{Leb}[0,1]$ for $i = 1,2,3$ exists if and only if the three mid-level lines $\{u_i = \frac{1}{2}\}$ have a common point p. In that case $\sum \mathrm{rw} \ge 1$ for every covering of Δ by planks parallel to these directions, and exactly one of the following holds:*

(a) *the full edges are pairwise distinct: up to flips the triple is cyclic, p lies in the open medial triangle, and Theorems 6.3 and 6.6 apply verbatim (explicit witness μ_p; unique tight covering);*

(b) *all three directions have the same full edge e: then p is the midpoint of e, the uniform measure on e is a witness, and the bound follows from covering e alone (the trivial one-dimensional case recorded in Theorem 2.5).*

Proof. Necessity. If μ exists, its barycenter $p \in \Delta$ satisfies $u_i(p) = \mathbb{E}_\mu[u_i] = \frac{1}{2}$ for every i (no invertibility of V is needed), so the mid-lines concur at p.

Classification. (1) If the three full edges are distinct, flipping each direction so that its $(0,1)$-pair is oriented cyclically makes the triple cyclic; by Corollary 6.4, concurrence then places p in the open medial triangle and yields μ_p, and Theorem 6.6 the unique tight covering. (2) Suppose exactly two directions, say u_i and u_j, share their full edge e, and let x_v be the barycentric coordinate of the vertex v opposite e. After a flip, u_i and u_j agree on e, so $u_i - u_j = (\tau_i - \tau_j)x_v$ with $\tau_i \ne \tau_j$ (equality would make them parallel). Hence $\{u_i = \frac{1}{2}\} \cap \{u_j = \frac{1}{2}\} \subset \{x_v = 0\}$, the line through e, on which both mid-lines pass only through the midpoint of e; but the third direction is not full on e, its values at the endpoints of e being $\{0, \tau_k\}$ or $\{\tau_k, 1\}$, so its value at the midpoint is $\frac{\tau_k}{2}$ or $\frac{1+\tau_k}{2}$, never $\frac{1}{2}$. Thus the mid-lines of such a triple *never* concur; consistently, no witness measure exists ($\mathbb{E}_\mu[u_i] = \mathbb{E}_\mu[u_j] = \frac{1}{2}$ would force $\mathbb{E}_\mu[x_v] = 0$, hence $\operatorname{supp}\mu \subset e$, on which u_k is not onto $[0,1]$). This shape contributes to neither side of the equivalence. (3) If all three directions share the full edge e, then after flips each restricts to e as the identity parameter, so all mid-lines pass through the midpoint of e (concurrence always holds), and the uniform probability measure on e has all three pushforwards equal to $\mathrm{Leb}[0,1]$. For the bound, a covering of Δ covers e, on which the i-th plank traces an interval of length r_i; hence $\sum r_i \ge 1$. ∎

Remark 6.12. Theorem 6.11 answers the existence question of [5] completely — but *only* — for three directions on the triangle. In particular, *within the uniform-marginal witness-measure* *approach*, the remaining three-direction case is that of non-concurrent triples, for which no witness measure can exist. This is a statement about the reach of the method, not a reduction of Conjecture 1.1 on the triangle (which concerns arbitrarily many planks in arbitrary directions).

The trichotomy of Theorem 6.11 in one glance:

full-edge pattern	concurrence	witness support	bound mechanism
three distinct edges (cyclic, mod flips)	iff mid-lines concur (two-parameter family)	weighted perimeter μ_p on $\partial\Delta$	transport, $D = 1$, tight and rigid
exactly two coincide	never	none exists	transport, $D > 1$: $\sum \mathrm{rw} \geq 1/D$ only
all three coincide, edge e	always (midpoint of e)	uniform measure on e	one-dimensional reduction

Rigidity (Theorem 6.6) concerns the concurrent members of the cyclic family. The same canonical configuration, however, *covers* for every cyclic parameter, with an excess that measures non-concurrence exactly.

Theorem 6.13 (canonical covering of an arbitrary cyclic triple)**.** *Let $\tau \in (0, 1)^3$ be any cyclic parameter (no concurrence assumed), and let λ_i, ρ_i be as in Proposition A.1. Write $\Pi = \tau_1\tau_2\tau_3$ and $Q = (1 - \tau_1)(1 - \tau_2)(1 - \tau_3)$. Then the planks $P_i = \{u_i \in [\lambda_i, \rho_i]\}$ cover Δ, with*

$$\sum_i \mathrm{rw}(P_i) \;=\; 1 + \frac{(\Pi - Q)^2}{(1 + \Pi)(1 + Q)}.$$

The excess vanishes if and only if $\Pi = Q$, which for cyclic triples is equivalent to concurrence of the mid-lines; in the concurrent case the configuration is the unique tight covering of Theorem 6.6.

Proof. Set $g_i = 1 - \tau_{i+1}(1 - \tau_{i+2}) \in (0, 1)$ (indices mod 3), so that $\lambda_i = \tau_i g_i/(1 + \Pi)$ and $\rho_i = g_{i+1}/(1 + Q)$ in the formulas of Proposition A.1. The affine dependency of the triple (Proposition 7.9(i)) has the explicit positive form

$$\sum_i c_i u_i \equiv 1, \qquad c_i = \frac{g_i}{1 + \Pi} > 0 :$$

at the vertex A, $c_1\tau_1 + c_3 = \big(\tau_1 - \tau_1\tau_2 + \Pi + 1 - \tau_1 + \tau_1\tau_2\big)/(1 + \Pi) = 1$, and similarly at B and C. Three polynomial identities, each a direct expansion (they are also verified symbolically in the ancillary files):

$$(1 + \Pi)^2 - \sum_i \tau_i g_i^2 = g_1g_2g_3, \qquad \sum_i g_i g_{i+1} - (1 + \Pi)(1 + Q) = g_1g_2g_3,$$

$$(1 + \Pi)\sum_i g_i - (1 + Q)\sum_i \tau_i g_i - (1 + \Pi)(1 + Q) = (\Pi - Q)^2.$$

The first two give the strict *margins*

$$\sum_i c_i\lambda_i = 1 - \frac{g_1g_2g_3}{(1 + \Pi)^2} < 1, \qquad \sum_i c_i\rho_i = 1 + \frac{g_1g_2g_3}{(1 + \Pi)(1 + Q)} > 1,$$

and the third is the displayed excess formula, since $\sum \mathrm{rw} = \sum_i \rho_i - \sum_i \lambda_i = \big(\sum g_i\big)/(1 + Q) - \big(\sum \tau_i g_i\big)/(1 + \Pi)$.

Coverage. By Proposition A.1 the traces of the P_i tile all three edges, so $\partial\Delta \subset \bigcup P_i$. The proof of Lemma 6.8 now applies verbatim with (a_i, S) replaced by $(c_i, 1)$: the boundary-contact count and the Pocket Lemma are unchanged; mixed sign patterns force their three level lines to concur and are excluded by boundedness; and the patterns (lo, lo, lo), (hi, hi, hi) would force $1 \le \sum c_i\lambda_i$, resp. $1 \ge \sum c_i\rho_i$ — contradicting the margins.

The excess and concurrence. From $c_i = g_i/(1+\Pi)$,

$$\textstyle\sum_i c_i = \frac{2+\Pi+Q}{1+\Pi}, \qquad 1 - \frac{1}{2}\sum_i c_i = \frac{\Pi-Q}{2(1+\Pi)},$$

using $\sum_i g_i = 3 - \left(\sum \tau_i - \sum \tau_i\tau_{i+1}\right) = 2 + \Pi + Q$. So the concurrence defect of §7 has the closed form $\delta_c(u_\tau) = |\Pi - Q| \,/\, \big(2(2 + \Pi + Q)\big)$, and $\Pi = Q$ iff $\delta_c = 0$ iff the mid-lines concur — necessarily inside Δ (Lemma 7.10, proved independently in §7) — iff $D(u_\tau) = 1$. At $\Pi = Q$ the excess vanishes and the configuration is the tight covering of Theorem 6.6. ☐

Remark 6.14. Theorem 6.13 extends the sharp half of Theorem 6.6 quantitatively beyond the concurrent locus: *every* cyclic triple carries a canonical non-degenerate covering whose excess over 1 is exactly the *concurrence penalty* $(\Pi-Q)^2/\big((1+\Pi)(1+Q)\big)$. For the sandwich parameter $\tau = (\frac{13}{25}, \frac{1}{2}, \frac{1}{2})$ of Remark 7.20 the excess is $\frac{1}{12656}$. In the language of Definition 2.3 this covering has total width 1+penalty > 1, so it does *not* improve the trivial bound $C^{111}_\Delta(u_\tau) \le C_\Delta(u_\tau) \le 1$; its value is structural — it exhibits the natural covering family interpolating the rigid tight coverings, and it feeds the covering-constant program of §10.

The concurrence condition is dimension-agnostic: the cyclic medians of Δ^d ($m_i(e_i) = \frac{1}{2}$, $m_i(e_{i+1}) = 0$, $m_i(e_{i-1}) = 1$, else $\frac{1}{2}$) have row-sums $\frac{d+1}{2}$, hence concur at the centroid in every dimension. (A caution specific to $d = 3$: there $m_{i+2} = 1 - m_i$, so this literal family collapses to two directions up to flips; this happens exactly when $d+1$ divides 4.) Section 8 constructs witness measures for a non-degenerate one-parameter family of concurrent quadruples on Δ^3, extending the transport method beyond the plane; for $d \ge 4$ the question is open (Problem 10.11).

7. Duality and exact values for the transport defect

Theorem 6.11 says the transport method certifies the full bound $\sum \mathrm{rw} \ge 1$ exactly when the mid-lines concur. In the language of §2:

Corollary 7.1. *For a pairwise non-parallel triple u on Δ, $D(u) = 1$ if and only if the three mid-level lines concur; otherwise $D(u) > 1$.*

Proof. Combine Proposition 2.2(iii) with Theorem 6.11. ☐

This section quantifies how the method degrades off the concurrent locus, and develops the tool that governs the question: a strong linear-programming duality for D_K, with explicit certificates on both sides.

The concurrence defect and the moment bound. The necessary condition of Proposition 6.1 can be made quantitative. For a family $u = (u_1, \dots, u_n)$ of directions on a convex body K define the *concurrence defect*

$$\delta(u) \;=\; \min_{p\in K}\; \max_i \left|u_i(p) - \tfrac{1}{2}\right|, \tag{2}$$

a minimum of a continuous function on a compact set, hence attained; clearly $\delta(u) = 0$ if and only if the mid-level sets $\{u_i = \frac{1}{2}\}$ have a common point in K. Moreover $\delta(u) < \frac{1}{2}$: at any interior p each $u_i(p) \in (0, 1)$, since an affine function on K attaining its minimum 0 at an interior point would be identically 0.

Proposition 7.2 (moment bound)**.** *For every family u of directions on a convex body K, $D_K(u) \ge \big(1 - 2\delta(u)\big)^{-1}$. In particular, for the facet triple $u = (\lambda_1, \lambda_2, \lambda_3)$ on the triangle one has $\delta = \frac{1}{6}$ and $D \ge \frac{3}{2}$.*

Proof. Let ν be a probability measure on $[0,1]$ with $\nu \le D\,\mathrm{Leb}$ and distribution function $F(t) \le \min(1, Dt)$. Its mean is $\int_0^1 (1-F) \ge \int_0^1 \max(0, 1-Dt)\,dt = \frac{1}{2D}$, and by the mirror argument the mean lies in $[\frac{1}{2D}, 1-\frac{1}{2D}]$. If μ is admissible for D and p is its barycenter, then $p \in K$ and the mean of $u_{i\#}\mu$ is $u_i(p)$, so $\delta(u) \le \max_i |u_i(p) - \frac{1}{2}| \le \frac{1}{2} - \frac{1}{2D}$, which rearranges to the claim. For the facets, $\sum_i \lambda_i(p) = 1$ forces some $\lambda_i(p) \le \frac{1}{3}$, so $\delta \ge \frac{1}{6}$; the centroid attains $\frac{1}{6}$. ☐

On the triangle the concurrence defect — of *any* family, of any size — can never exceed the facet value:

Theorem 7.3 (centroid bound)**.** *For every finite family $u_1, \dots, u_n$ of directions on the triangle, $\delta(u) \le \frac{1}{6}$, witnessed by the centroid G; the facet triple attains $\frac{1}{6}$. Consequently the moment bound never certifies $D(u) > \frac{3}{2}$ on the triangle, for any number of directions.*

Proof. A direction u has vertex values $\{0, s, 1\}$ in some order, $s \in [0,1]$, so $u(G) = \frac{0+s+1}{3} = \frac{1+s}{3} \in [\frac{1}{3}, \frac{2}{3}]$, i.e. $|u(G) - \frac{1}{2}| \le \frac{1}{6}$, for *every* direction simultaneously. For the facet triple $\delta = \frac{1}{6}$ by Proposition 7.2. (Equality $|u(G) - \frac{1}{2}| = \frac{1}{6}$ forces $s \in \{0,1\}$, i.e. two equal vertex values, as for facet directions.) ☐

Among genuine triples, the extremal case is unique:

Proposition 7.4 (uniqueness of the extremal triple)**.** *A pairwise non-parallel triple on Δ has $\delta(u) = \frac{1}{6}$ if and only if it is the facet triple modulo flips $u \mapsto 1-u$.*

Proof. Flips of facet directions give $\delta = \frac{1}{6}$ by Proposition 7.2 (δ is flip-invariant). Conversely, suppose $\delta(u) = \frac{1}{6}$. Since $\delta \le \max_i |u_i(G) - \frac{1}{2}| \le \frac{1}{6}$, the centroid is a minimizer with value exactly $\frac{1}{6}$. Let $B = \{i : |u_i(G) - \frac{1}{2}| = \frac{1}{6}\}$ $(\ne \varnothing)$; by the equality analysis in Theorem 7.3, each $i \in B$ has $s_i \in \{0,1\}$, i.e. $u_i \in \{x_v, 1-x_v\}$ for some vertex $v = v(i)$, where x_v denotes the barycentric coordinate of v; in either case the *improving gradient* at G (the direction of decrease of $|u_i - \frac{1}{2}|$) is $g_i = \nabla x_{v(i)}$. If some vector d in the plane of Δ had $\langle g_i, d\rangle > 0$ for all $i \in B$, then for small $t > 0$ the interior points $G + td$ would satisfy $|u_i - \frac{1}{2}| < \frac{1}{6}$ for $i \in B$ (strict improvement) and for $i \notin B$ (continuity), contradicting $\delta = \frac{1}{6}$. By Gordan's alternative, 0 lies in the convex hull of $\{g_i : i \in B\}$. Now $|B| = 1$ is impossible ($g_i \ne 0$), and $|B| = 2$ would force $g_i \parallel g_j$, i.e. $u_i \parallel u_j$ — excluded. So $|B| = 3$: all three directions are of the form x_v or $1 - x_v$, and pairwise non-parallelism forces three *distinct* vertices: the facet triple modulo flips. ☐

At the facets the moment bound is exact:

Theorem 7.5. *For the facet triple, $D(\lambda_1, \lambda_2, \lambda_3) = \frac{3}{2}$. The infimum is attained by the uniform probability measure on the perimeter of the inscribed triangle with vertices*

$$P_0 = (0, \tfrac{1}{3}, \tfrac{2}{3}), \qquad P_1 = (\tfrac{2}{3}, 0, \tfrac{1}{3}), \qquad P_2 = (\tfrac{1}{3}, \tfrac{2}{3}, 0),$$

whose three marginals all equal $\frac{3}{2} \cdot \mathbf{1}_{[0,2/3]}$.

Proof. The lower bound is Proposition 7.2. For the upper bound give each side of the loop mass $\frac{1}{3}$, uniformly in the parameter. The coordinate λ_1 is affine on each side, with values $0 \to \frac{2}{3}$ on P_0P_1 (density $\frac{1}{2}$ on $[0, \frac{2}{3}]$), $\frac{2}{3} \to \frac{1}{3}$ on P_1P_2 (density 1 on $[\frac{1}{3}, \frac{2}{3}]$), and $\frac{1}{3} \to 0$ on P_2P_0 (density 1 on $[0, \frac{1}{3}]$); the total is $\frac{3}{2}$ on $[0, \frac{2}{3}]$. The loop is invariant under the cyclic rotation of Δ, so the same holds for λ_2, λ_3. ☐

Remark 7.6. Theorem 7.5 makes precise the complementarity of the two methods in this paper: for facet directions the tiling argument of Theorem 3.3 certifies the sharp bound 1, while the transport method cannot certify more than $1/D = \frac{2}{3}$ — and now provably certifies exactly that.

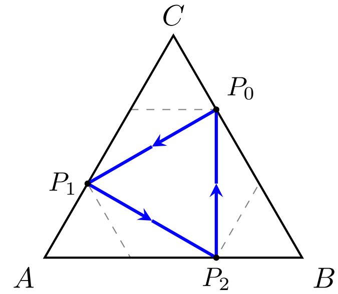

FIGURE 3. The witness of Theorem 7.5: the uniform measure on the inscribed loop $P_0P_1P_2$ (mass $\frac{1}{3}$ per side) has all three marginals equal to $\frac{3}{2} \cdot \mathbf{1}_{[0,2/3]}$, so $D = \frac{3}{2}$ for the facet triple. Dashed: the hexagon $\{\max_i \lambda_i \le \frac{2}{3}\}$, whose alternate vertices the loop connects.

Conversely, for the concurrent cyclic triples the transport bound is sharp (Theorems 6.3 and 6.6) where no facet tiling is available.

Duality. D_K is the value of an infinite-dimensional linear program, and it has a concrete dual. Call a tuple $\psi = (\psi_1, \dots, \psi_n)$ of continuous functions on $[0,1]$ *admissible* if $\psi_i \ge 0$ for all i and $\sum_i \int_0^1 \psi_i(t)\,dt = 1$.

Theorem 7.7 (minimax duality)**.** *For every convex body K and every family $u_1, \dots, u_n$ of directions on K,*

$$D_K(u) \;=\; \sup_{\psi \text{ admissible}} \; \min_{x \in K} \sum_{i=1}^{n} \psi_i\big(u_i(x)\big).$$

Weak duality is elementary and two-sided: every D-admissible measure certifies $D_K(u) \le D$, and every admissible ψ certifies $D_K(u) \ge \min_{x\in K} \sum_i \psi_i(u_i(x))$. The supremum is unchanged if the ψ_i range over bounded Borel functions. We do not *assert that the supremum is attained in general; it is attained in every exact example of this paper.*

Proof. Weak duality. If μ is D-admissible and ψ admissible, then

$$\min_{x\in K} \sum_i \psi_i(u_i(x)) \;\le\; \int_K \sum_i \psi_i(u_i(x))\,d\mu(x) \;=\; \sum_i \int_0^1 \psi_i\,d\nu_i \;\le\; D \sum_i \int_0^1 \psi_i\,dt \;=\; D,$$

where $\nu_i := u_{i\#}\mu$; the middle inequality holds because $\nu_i \le D \cdot \mathrm{Leb}$ and $\psi_i \ge 0$. Since the infimum defining $D_K(u)$ is attained (Proposition 2.2(i)), the supremum is at most $D_K(u)$; the same computation applies verbatim to bounded Borel ψ_i, so enlarging the class cannot raise the supremum past $D_K(u)$.

The pairing. For $\mu \in \mathcal{P}(K)$ and admissible ψ put $\Phi(\psi,\mu) = \sum_i \int_K \psi_i(u_i(x))\,d\mu(x)$, a bilinear functional.

Claim 1: for fixed μ, $\sup_\psi \Phi(\psi,\mu) = D_{\min}(\mu) := \inf\{D : \nu_i \le D \cdot \mathrm{Leb}\ \forall i\} \in [1,\infty]$. The inequality $\le$ is the computation above. For $\ge$, let $1 \le D < D_{\min}(\mu)$; then $\nu_i(A) > D|A|$ for some i and some Borel $A \subseteq [0,1]$. By inner regularity of ν_i choose a compact $F \subseteq A$ with $\nu_i(F) > D|A| \ge D|F|$; by outer regularity of Lebesgue measure choose $U \supseteq F$, relatively open in $[0,1]$, with $D|U| < \nu_i(F)$. By Urysohn's lemma there is a continuous g with $\mathbf{1}_F \le g \le \mathbf{1}_U$; then $F \ne \varnothing$ (as $\nu_i(F) > 0$) forces $\int_0^1 g\,dt > 0$, and

$$\int_0^1 g\,d\nu_i \;\ge\; \nu_i(F) \;>\; D|U| \;\ge\; D\int_0^1 g\,dt.$$

The tuple with $\psi_i = g/\int g\,dt$ and $\psi_j = 0$ for $j \neq i$ is admissible and gives $\Phi > D$. Hence $\sup_\psi \Phi(\psi,\mu) \geq D_{\min}(\mu)$ (for $D < 1$ use $\psi_1 \equiv 1$, which gives $\Phi = 1$). By Definition 2.1, $D_K(u) = \inf_\mu D_{\min}(\mu)$.

Claim 2: for fixed admissible ψ, $\inf_{\mu\in\mathcal{P}(K)} \Phi(\psi,\mu) = \min_{x\in K} \sum_i \psi_i(u_i(x))$: the integrand $x \mapsto \sum_i \psi_i(u_i(x))$ is continuous on the compact K, so the minimum is attained; every μ-average dominates it, and Dirac measures realize it.

Minimax. $\mathcal{P}(K)$ with the weak-$*$ topology is a compact convex subset of $C(K)^*$, and the admissible ψ form a convex subset of the Banach space $C([0,1])^n$. Φ is bilinear, weak-$*$ continuous in μ for fixed ψ (the integrand is continuous on K), and sup-norm continuous in ψ for fixed μ. Sion's minimax theorem [6] gives

$$\inf_{\mu\in\mathcal{P}(K)} \sup_{\psi} \Phi(\psi,\mu) = \sup_{\psi} \inf_{\mu\in\mathcal{P}(K)} \Phi(\psi,\mu).$$

By Claim 1 the left side is $\inf_\mu D_{\min}(\mu) = D_K(u)$; by Claim 2 the right side is the asserted supremum. □

Corollary 7.8 (certificates)**.** *Upper bounds for $D_K(u)$ are certified by explicit measures, lower bounds by explicit admissible ψ: verifying a lower bound B requires only the normalization $\sum_i \int \psi_i = 1$ and the pointwise inequality $\sum_i \psi_i(u_i(x)) \geq B$ on K — no optimization. By Theorem 7.7 the two sides meet at $D_K(u)$; every exact defect value in this paper is certified by such a matched pair. (Lower bounds for D quantify what the transport method* cannot *certify; upper bounds for D yield covering bounds via Proposition 2.2(ii).)*

For the facet triple the dual side is three lines: $\psi_i(t) = (1 - \frac{3}{2}t)_+$ for $i = 1,2,3$ is admissible (each integral equals $\frac{1}{3}$), and pointwise $\psi_i(\lambda_i) \geq 1 - \frac{3}{2}\lambda_i$, so on Δ

$$\textstyle\sum_i \psi_i(\lambda_i) \;\geq\; 3 - \frac{3}{2}\sum_i \lambda_i \;=\; \frac{3}{2},$$

certifying $D \geq \frac{3}{2}$ and matching the loop measure of Theorem 7.5 exactly. Note where equality lives: the sum equals $\frac{3}{2}$ precisely on the hexagon $\{\max_i \lambda_i \leq \frac{2}{3}\}$, which carries the inscribed loop, and each ψ_i vanishes precisely on $[\frac{2}{3}, 1]$, where the slack $\rho_i = \frac{3}{2}\mathbf{1}_{[2/3,1]}$ of the loop's marginal lives — the equality pattern of Proposition 7.13 below in the flesh (Figure 4).

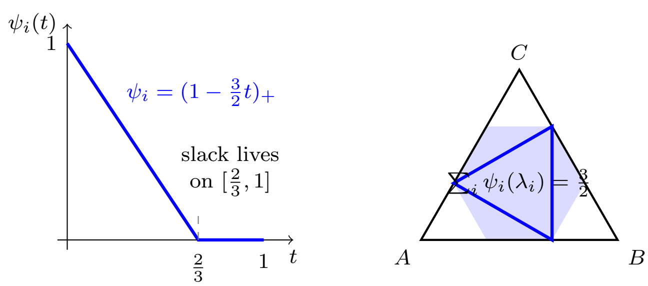

Figure 4. The matched primal–dual pair at the facet triple. Left: the dual certificate $\psi_i(t) = (1 - \frac{3}{2}t)_+$ (one wedge per direction, total integral 1), which vanishes exactly where the slack of the optimal marginals lives. Right: the contact set $\{\sum_i \psi_i(\lambda_i) = \frac{3}{2}\}$ is the shaded hexagon $\{\max_i \lambda_i \leq \frac{2}{3}\}$, and the optimal measure — the inscribed loop of Theorem 7.5 — is supported inside it: the equality pattern of Proposition 7.13.

The moment bound, too, has a dual incarnation: an explicit family of *wedge* certificates built from the affine dependency of the triple.

Proposition 7.9 (wedge certificates). *Let $u = (u_1, u_2, u_3)$ be pairwise non-parallel directions on Δ.*

(i) *There is an affine dependency $\sum_i c_i u_i \equiv c_0$ with $c = (c_1, c_2, c_3)$, unique up to scale, and all $c_i \neq 0$. (The constant may vanish: $c_0 = 0$ occurs exactly for the degenerate families all of whose u_i vanish at a common point of the plane, e.g. triples sharing one full edge; every quantity below is homogeneous of degree 0 in (c, c_0), so no normalization is assumed.) Set $A = \sum_i |c_i|$ and*

$$\kappa_1 = c_0 - \sum_{c_i<0} c_i, \qquad \kappa_2 = \sum_{c_i>0} c_i - c_0,$$

so $\kappa_1 + \kappa_2 = A$. Then $\kappa_1, \kappa_2 > 0$, and with $\kappa = \min(\kappa_1, \kappa_2)$ and $\theta = 2\kappa/A \in (0, 1]$,

$$D(u) \ \geq \ \frac{A}{2\kappa} \ = \ \frac{1}{1 - 2\delta_c(u)}, \qquad \delta_c(u) := \frac{\left|c_0 - \frac{1}{2}\sum_i c_i\right|}{A},$$

certified by the admissible wedges $\psi_i(t) = \frac{2|c_i|}{A\theta^2}\big(\theta - \varphi_i(t)\big)_+$, where $\varphi_i(t) = t$ for $c_i > 0$ and $\varphi_i(t) = 1 - t$ for $c_i < 0$ if $\kappa = \kappa_1$, and with $t \leftrightarrow 1 - t$ exchanged if $\kappa = \kappa_2$.

(ii) *$\delta_c(u)$ is the ℓ^∞-distance from the center $z = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ to the plane $\Pi = \{t \in \mathbb{R}^3 : \sum_i c_i t_i = c_0\}$, which contains the image triangle $T_u = \{(u_1(x), u_2(x), u_3(x)) : x \in \Delta\}$. Hence $\delta_c(u) \leq \delta(u)$: the wedge bound is at most the moment bound.*

(iii) *If $\delta_c > 0$, the point of Π nearest to z in ℓ^∞ is unique, namely p^* with $p^*_i = \frac{1}{2} + \epsilon\, \delta_c \operatorname{sign}(c_i)$, $\epsilon = \operatorname{sign}\big(c_0 - \frac{1}{2}\sum_i c_i\big)$; and $\delta_c(u) = \delta(u)$ — i.e. the wedge certificate recovers the moment bound exactly — if and only if $p^* \in T_u$. Theorem 7.11 below shows that this criterion is* always *satisfied.*

Proof. (i) Affine functions on the plane of Δ form a three-dimensional space, so $u_1, u_2, u_3, \mathbf{1}$ are linearly dependent: $\sum c_i u_i = c_0 \mathbf{1}$ with $(c, c_0) \neq 0$, and $c = 0$ would force $c_0 = 0$. If some $c_i = 0$, the relation makes the other two directions parallel — excluded; a second, independent relation would eliminate one u_i and make the remaining two parallel or constant — also excluded. So c is unique up to scale with all $c_i \neq 0$.

Evaluating the dependency at any $x \in \Delta$,

$$\kappa_1 = c_0 - \sum_{c_i<0} c_i = \sum_{c_i>0} c_i u_i(x) + \sum_{c_i<0} |c_i|\big(1 - u_i(x)\big) \ \geq \ 0,$$

a *constant* nonnegative combination; if it vanished, each summand would vanish for every x, making some u_i constant ($\equiv 0$ or $\equiv 1$), contradicting ontoness. So $\kappa_1 > 0$, and symmetrically $\kappa_2 > 0$; $\kappa_1 + \kappa_2 = A$ is immediate, whence $\theta = 2\kappa/A \leq 1$.

Say $\kappa = \kappa_1$ (else exchange $t \leftrightarrow 1 - t$ throughout, which replaces the telescoped constant below by κ_2). Since $\int_0^1 (\theta - \varphi_i(t))_+ \, dt = \theta^2/2$ for either orientation of φ_i, the stated ψ is admissible: $\sum_i \int \psi_i = \frac{2}{A\theta^2} \cdot A \cdot \frac{\theta^2}{2} = 1$. Pointwise $(\theta - \varphi)_+ \geq \theta - \varphi$ and, by the display above, $\sum_i |c_i| \varphi_i(u_i(x)) = \kappa_1$ for every x; hence

$$\sum_i \psi_i(u_i(x)) \ \geq \ \frac{2}{A\theta^2}\Big(A\theta - \sum_i |c_i| \varphi_i(u_i(x))\Big) = \frac{2(A\theta - \kappa_1)}{A\theta^2} = \frac{A}{2\kappa_1}$$

for $\theta = 2\kappa_1/A$. Weak duality (Theorem 7.7) gives $D(u) \geq A/(2\kappa)$. Finally $c_0 - \frac{1}{2}\sum c_i = \kappa_1 - \frac{A}{2} = \frac{A}{2} - \kappa_2$, so $|c_0 - \frac{1}{2}\sum c_i| = \frac{A}{2} - \kappa$ and $1 - 2\delta_c = 2\kappa/A$.

(ii) The ℓ^∞-distance from a point z to the plane $\{\langle c, t\rangle = c_0\}$ is $|\langle c, z\rangle - c_0|/\|c\|_1$ (dual norms), which at the center is δ_c. $T_u \subset \Pi$ by the dependency, and $\delta(u) = \min_{t \in T_u} \|t - z\|_\infty \geq \operatorname{dist}_\infty(z, \Pi) = \delta_c$.

(iii) For $t \in \Pi$ with $\|t-z\|_\infty = \delta_c$: $|\langle c, t-z\rangle| = |c_0 - \langle c, z\rangle| = \delta_c A$ while $|\langle c, t-z\rangle| \le \|c\|_1 \|t-z\|_\infty = \delta_c A$, with equality only if $(t-z)_i = \epsilon\, \delta_c \operatorname{sign}(c_i)$ for all i (all $c_i \neq 0$): the nearest point is the single p^*. Then $\delta(u) = \delta_c$ iff some point of T_u realizes the plane distance iff $p^* \in T_u$. $\square$

The two defects in fact always coincide. The key is a counting lemma: a level set of a *common* value can only be shared by few directions at a point outside the triangle. (At the level $\frac{1}{2}$ this says mid-level lines of a genuine triple cannot concur outside Δ.)

Lemma 7.10 (counting lemma)**.** *Let q be a point of the plane of Δ with $q \notin \Delta$ (some barycentric coordinate negative; $\sum_j q_j = 1$), and let $a \in (0,1)$. Then at most* two *pairwise non-parallel directions take the value a at q. Consequently: if the three mid-level lines $\{u_i = \frac{1}{2}\}$ of a pairwise non-parallel triple have a common point q, then $q \in \Delta$; and $\delta_c(u) = 0 \iff \delta(u) = 0 \iff D(u) = 1$.*

Proof. The flip $u \mapsto 1-u$ bijects directions with value a at q onto directions with value $1-a$ at q and preserves parallelism, so we may assume $a \le \frac{1}{2}$. A direction with value 0 at vertex j_0, 1 at vertex j_1 and $s \in [0,1]$ at the remaining vertex j_s takes at q the value $q_{j_1} + s\, q_{j_s}$.

Case $a = \frac{1}{2}$. The value condition reads $q_{j_1} + s\, q_{j_s} = \frac{1}{2}$; since $q_{j_1} + q_{j_s} = 1 - q_{j_0}$, a solution $s \in [0,1]$ exists iff $\frac{1}{2}$ lies weakly between q_{j_1} and $1 - q_{j_0}$, iff $(q_{j_0} - \frac{1}{2})(q_{j_1} - \frac{1}{2}) \ge 0$. The ordered patterns (j_0, j_1) and (j_1, j_0) produce flipped, hence parallel, directions, so the count is at most the number of *unordered* pairs $\{j,k\}$ with $(q_j - \frac{1}{2})(q_k - \frac{1}{2}) \ge 0$. All three pairs qualify only when all $q_j \le \frac{1}{2}$ (all $\ge \frac{1}{2}$ contradicts $\sum q_j = 1$), and then $q_j = 1 - q_k - q_l \ge 0$ for each j: $q \in \Delta$, excluded. So the count is at most 2.

Case $a < \frac{1}{2}$. Write $r_j = q_j - a$ and $r'_j = q_j - (1-a)$, so $r_j - r'_j = 1 - 2a > 0$. First, $q_{j_s} = 0$ is impossible for a solvable pattern: it would force $q_{j_1} = a$ and $q_{j_0} = 1-a$, making all coordinates nonnegative — but $q \notin \Delta$. So each solvable pattern has the unique $s = (a - q_{j_1})/q_{j_s}$, and $s \in [0,1]$ iff a lies weakly between q_{j_1} and $1 - q_{j_0}$, iff $r_{j_1} r'_{j_0} \ge 0$.

Parallelism and collisions. Two directions are parallel iff one is the flip of the other; the flip of a solution takes value $1 - a \neq a$ at q, so distinct solutions are never parallel. A solution with $s \in (0,1)$ has three distinct vertex values, so its pattern is determined by the direction: distinct *strict* patterns give distinct directions. The boundary cases collide: $s = 0$ (iff $r_{j_1} = 0$) gives the coordinate direction x_{j_1} from *both* patterns with that j_1; $s = 1$ (iff $r'_{j_0} = 0$) gives $1 - x_{j_0}$ from both patterns with that j_0. Hence the number N of distinct pairwise non-parallel directions with value a at q is

$$N = \#\{\text{solvable patterns with strict signs}\} + \#\{j : r_j = 0\} + \#\{j : r'_j = 0\},$$

the two boundary counts contributing one direction each (x_j, resp. $1 - x_j$; these are pairwise non-parallel for distinct indices, and $r_j = r'_j = 0$ is impossible).

Counting under externality. Fix m with $q_m < 0$; then $r_m, r'_m < 0$ strictly. Let $\{j,k\}$ be the other two indices, so $q_j + q_k = 1 - q_m > 1$. A pattern (j_0, j_1) is strict iff ($r'_{j_0} > 0$ and $r_{j_1} > 0$) or ($r'_{j_0} < 0$ and $r_{j_1} < 0$). Three cases.

(1) $r_j, r_k \le 0$*:* then $q_j + q_k \le 2a < 1$, impossible.

(2) *Exactly one positive, say $r_j > 0 \ge r_k$:* then $q_j = 1 - q_m - q_k > 1 - a$, so $r'_j > 0$ strictly, and $r'_k = r_k - (1-2a) < 0$ strictly. Positive-side strict patterns need $r_{j_1} > 0$ with $j_1 \neq j_0 = j$: none. Negative-side strict patterns have $j_0, j_1 \in \{k, m\}$: if $r_k < 0$, the two patterns $(k,m), (m,k)$ and no boundary terms; if $r_k = 0$, the single pattern (k,m) plus the boundary direction x_k. Either way $N = 2$.

(3) $r_j, r_k > 0$*:* sub-classify by the signs of (r'_j, r'_k). Both > 0: strict patterns $(j,k), (k,j)$ only (a negative-side pattern would need $j_0 = j_1 = m$); $N = 2$. Signs $(+,-)$: strict patterns

(j,k) and (k,m); $N=2$. Signs $(+,0)$: strict (j,k), boundary $1-x_k$; $N=2$. Both <0: strict $(j,m),(k,m)$; $N=2$. Signs $(0,-)$: strict (k,m), boundary $1-x_j$; $N=2$. Signs $(0,0)$: no strict patterns, boundaries $1-x_j, 1-x_k$; $N=2$.

In every feasible case $N \le 2$, proving the counting statement.

Consequences. If the three mid-lines concur at q, then three pairwise non-parallel directions take the value $\frac{1}{2}$ at q, so $q \in \Delta$. If $\delta_c = 0$, the center lies on Π (Proposition 7.9(ii)); the mid-lines of u_1, u_2 meet at a single point q, and $c_0 = \sum_i c_i u_i(q) = \frac{1}{2}(c_1+c_2) + c_3 u_3(q)$ together with $c_0 = \frac{1}{2}\sum c_i$ gives $u_3(q) = \frac{1}{2}$ ($c_3 \ne 0$): the mid-lines concur at $q \in \Delta$, so $\delta(u) = 0$ and $D(u) = 1$ (Corollary 7.1). The converses are trivial. ☐

Theorem 7.11 (the two defects coincide)**.** *For every pairwise non-parallel triple u on Δ,*

$$\delta(u) \;=\; \delta_c(u) \;=\; \frac{\left|c_0 - \frac{1}{2}\sum_i c_i\right|}{\sum_i |c_i|}\,.$$

Equivalently, the ℓ^∞-projection p^* of the center onto Π always lies in the image triangle T_u, and the single-knot wedge certificates of Proposition 7.9 always attain the moment bound. In particular δ is an explicit rational function of the vertex data; for cyclic triples, $\delta(u_\tau) = |\Pi - Q|/\big(2(2+\Pi+Q)\big)$ (Theorem 6.13).

Proof. By Proposition 7.9(ii) $\delta \ge \delta_c$, with equality iff $p^* \in T_u$ (part (iii)); the case $\delta_c = 0$ is Lemma 7.10. Assume $\delta_c > 0$ and, for contradiction, $p^* \notin T_u$.

Flip normalization. The flip $u_i \mapsto 1-u_i$ replaces $(c_i, c_0) \mapsto (-c_i, c_0 - c_i)$, leaves δ, δ_c invariant (both $c_0 - \frac{1}{2}\sum c$ and $\sum |c|$ are unchanged), and conjugates T_u and p^* by the reflection $t_i \mapsto 1-t_i$ of the i-th coordinate. Flipping every i with $c_i < 0$, we may assume all $c_i > 0$; then $p^* = (a,a,a)$ with $a = \frac{1}{2} + \epsilon\delta_c \in (0,1)$ (Proposition 7.9(iii), $\delta_c < \frac{1}{2}$).

Pulling back. The evaluation map $E(x) = (u_1(x), u_2(x), u_3(x))$ is an affine bijection from the plane of Δ onto Π: it is injective because u_1, u_2 are non-parallel, hence affine coordinates, and its image is a 2-plane contained in Π, hence equal to it. Let $q = E^{-1}(p^*)$: a point of the plane with $u_i(q) = a$ for $i = 1,2,3$, and $p^* \notin T_u = E(\Delta)$ means $q \notin \Delta$.

Conclusion. The three directions u_1, u_2, u_3 are pairwise non-parallel and all take the value $a \in (0,1)$ at $q \notin \Delta$ — contradicting Lemma 7.10. Hence $p^* \in T_u$ and $\delta = \delta_c$. ☐

Remark 7.12. Theorem 7.11 collapses the a priori hierarchy: for every pairwise non-parallel triple,

$$\frac{1}{1-2\delta_c(u)} \;=\; \frac{1}{1-2\delta(u)} \;\le\; D(u):$$

the moment bound *is* the value of the best single-knot wedge pair, in closed form. The remaining inequality is strict in general: explicit multi-knot certificates beat the moment bound at explicit triples (Remark 7.20), and by Theorem 7.7 richer ψ are the *only* possible source of better lower bounds. For the facet triple both defects equal $\frac{1}{6}$ and the wedge certificate ($c = (1,1,1)$, $c_0 = 1$, $\theta = \frac{2}{3}$) is precisely the facet ψ displayed above; for the sandwich triple both equal $\frac{1}{450}$.

Proposition 7.13 (equality pattern)**.** *Let μ be a D-admissible measure and ψ an admissible tuple with* $\min_{x\in K}\sum_i \psi_i(u_i(x)) = D$*. Then $D = D_K(u)$, both are optimal, and:*

(i) *μ is supported on the* contact set $C_\psi = \big\{x \in K : \sum_i \psi_i(u_i(x)) = D\big\}$*;*
(ii) *for each i, $\int \psi_i \, d\rho_i = 0$, where $\rho_i = D \cdot \mathrm{Leb} - u_{i\#}\mu \ge 0$ is the slack (i.e. ψ_i vanishes ρ_i-a.e.).*

Conversely, if a D-admissible μ and an admissible ψ satisfy (ii) and μ is supported on the set where $\sum_i \psi_i \circ u_i$ attains its minimum over K, then $D = D_K(u)$ and both are optimal.

Proof. The weak-duality chain

$$D = \min_{x\in K} \sum_i \psi_i(u_i(x)) \ \le\ \int_K \sum_i \psi_i(u_i)\, d\mu \ =\ D\sum_i \int \psi_i\, dt - \sum_i \int \psi_i\, d\rho_i \ \le\ D$$

collapses to equalities. Equality on the left forces the continuous integrand $\sum_i \psi_i \circ u_i$ (which is $\ge D$ everywhere) to equal D on $\operatorname{supp}\mu$: (i). Equality on the right forces $\sum_i \int \psi_i\, d\rho_i = 0$ with nonnegative summands: (ii). Optimality: μ certifies $D_K(u) \le D$ and ψ certifies $D_K(u) \ge D$. For the converse, let $c = \min_{x\in K} \sum_i \psi_i(u_i(x))$; the support condition gives $\int \sum_i \psi_i(u_i)\, d\mu = c$, while (ii) gives $\int \sum_i \psi_i(u_i)\, d\mu = D\sum_i \int \psi_i = D$; so $c = D$ and the first part applies. □

Remark 7.14 (skeleton measures and where the optimum lives). Let $D_\partial(u)$ denote the infimum of Definition 2.1 restricted to measures supported on $\partial\Delta$; always $D(u) \le D_\partial(u)$. On the concurrent locus $D_\partial = D = 1$: the witness μ_p of Theorem 6.3 lives on $\partial\Delta$. At the facet triple, by contrast, $D_\partial = +\infty$: any probability measure on $\partial\Delta$ gives positive mass to some closed edge, on which the opposite barycentric coordinate vanishes identically, so some marginal has an atom at 0 and no level D is feasible — the optimum abandons the boundary entirely (the loop of Theorem 7.5 is interior). For cyclic triples every direction is nonconstant on every edge, $D_\partial(u)$ is finite, and restricted to piecewise-uniform edge densities it is a finite linear program with rational data: its value is a certified upper bound (Corollary 7.8). By Proposition 7.13, $D_\partial(u) = D(u)$ exactly when some optimal measure is supported on $\partial\Delta$ — equivalently, when the contact set of some dual optimizer supports one. Whether that holds anywhere off the concurrent locus is open (Remark 7.20, Problem 10.12).

Stability near the concurrent locus. Near the concurrent locus the defect is small, with an explicit modulus; this gives bounds beyond the general constant of [4] on open sets of non-concurrent triples, for coverings in the corresponding direction families (see Remark 7.19).

Theorem 7.15. *Let $\tau^* \in (0,1)^3$ be a concurrent cyclic parameter with edge masses (α,β,γ) as in Theorem 6.3, and let $m = \min_i \min(\tau_i^*, 1-\tau_i^*)$. If $\tau \in (0,1)^3$ is a cyclic parameter with $\varepsilon := \max_i |\tau_i - \tau_i^*| < m$, then*

$$D(u_\tau) \ \le\ 1 + \frac{\varepsilon}{m(m-\varepsilon)}, \qquad \textit{hence} \qquad \sum \mathrm{rw} \ \ge\ \frac{m(m-\varepsilon)}{m(m-\varepsilon)+\varepsilon}$$

for every covering of Δ by planks parallel to the directions of τ.

Proof. Keep the *frozen* weights: let μ be the measure with masses (γ,α,β) on the edges (AB, BC, CA) — the mass convention of Theorem 6.3 — uniform in the parameter, and push it forward by the *perturbed* directions. For u_1 (vertex values $(\tau_1, 0, 1)$) the marginal density is $\alpha + \gamma/\tau_1$ on $[0,\tau_1]$ and $\alpha + \beta/(1-\tau_1)$ on $[\tau_1, 1]$; at τ_1^* both equal 1 by the choice of weights in Theorem 6.3. Subtracting,

$$\left(\alpha + \frac{\gamma}{\tau_1}\right) - 1 = \frac{\gamma\,(\tau_1^* - \tau_1)}{\tau_1\tau_1^*}, \qquad \left(\alpha + \frac{\beta}{1-\tau_1}\right) - 1 = \frac{\beta\,(\tau_1 - \tau_1^*)}{(1-\tau_1)(1-\tau_1^*)},$$

and since $\gamma,\beta \le 1$, $\tau_1^* \ge m$, $1-\tau_1^* \ge m$ and $\tau_1, 1-\tau_1 \ge m-\varepsilon$, both deviations are at most $\varepsilon/(m(m-\varepsilon))$ in absolute value; the same computation applies to u_2, u_3. Hence $u_{i\#}\mu \le (1 + \varepsilon/(m(m-\varepsilon)))\,\mathrm{Leb}$ for all i, and Proposition 2.2(ii) gives the covering bound: for $x = \varepsilon/(m(m-\varepsilon))$ the exact reciprocal is $1/(1+x) = m(m-\varepsilon)/\big(m(m-\varepsilon)+\varepsilon\big)$. □

Stability runs in both directions: a small defect forces near-concurrence, by our own moment bound.

Proposition 7.16 (inverse stability)**.** *If $D(u) \le 1+\eta$ for a family u on a convex body K, then* $\delta(u) \le \dfrac{\eta}{2(1+\eta)}$.

Proof. By Proposition 7.2, $(1-2\delta(u))^{-1} \le D(u) \le 1+\eta$, so $1-2\delta(u) \ge (1+\eta)^{-1}$, which rearranges to the claim. □

With Theorem 7.15 in one direction and Proposition 7.16 in the other, $D(u)-1$ is a two-sided quantitative measure of non-concurrence on the cyclic family: it vanishes exactly on the concurrent locus (Corollary 7.1) and is comparable to the concurrence defect on both sides.

Corollary 7.17. *Every covering of Δ by planks parallel to the directions of a cyclic parameter τ with $\max_i |\tau_i - \frac{1}{2}| \le \frac{1}{60}$ satisfies*

$$\sum \mathrm{rw} \;\ge\; \tfrac{29}{31} \;>\; 4\sqrt{3}-6.$$

Proof. Apply Theorem 7.15 with $\tau^* = (\frac{1}{2},\frac{1}{2},\frac{1}{2})$ (the medians, concurrent with $\alpha=\beta=\gamma=\frac{1}{3}$), $m=\frac{1}{2}$, $\varepsilon=\frac{1}{60}$: here $m(m-\varepsilon)=\frac{29}{120}$ and $m(m-\varepsilon)+\varepsilon=\frac{31}{120}$, so $D\le\frac{31}{29}$ and $\sum\mathrm{rw}\ge\frac{29}{31}$. The comparison $\frac{29}{31}>4\sqrt{3}-6$ is $29+6\cdot 31=215>124\sqrt{3}$, i.e. $215^2=46225>46128=48\cdot 31^2$. □

The medians play no special role: Theorem 7.15 certifies an explicit open neighbourhood of the *whole* concurrent family on which the transport bound exceeds the numerical value of $4\sqrt{3}-6$.

Corollary 7.18 (a certified neighbourhood of the concurrent locus)**.** *Set $k=\frac{2\sqrt{3}-3}{6}$, so that $1+k=\frac{1}{4\sqrt{3}-6}$. Let τ^* be any concurrent cyclic parameter and $m=\min_i\min(\tau_i^*,1-\tau_i^*)$. Then every cyclic parameter τ with*

$$\max_i |\tau_i-\tau_i^*| \;<\; \frac{km^2}{1+km}$$

satisfies $D(u_\tau)<\frac{1}{4\sqrt{3}-6}$, hence $\sum\mathrm{rw}>4\sqrt{3}-6$ for every covering of Δ by planks parallel to the directions of τ. The union of these boxes over the two-parameter concurrent family is an explicit open set of triples — almost all of them non-concurrent — with a transport bound past the constant of [4], subject to the caveat of Remark 7.19. It is the region certified by the upper bound of Theorem 7.15, not a description of the full sublevel set $\{D<\frac{1}{4\sqrt{3}-6}\}$, whose exact extent we do not know.

Proof. Put $\varepsilon=\max_i|\tau_i-\tau_i^*|$. From $\varepsilon<\frac{km^2}{1+km}<m$ (the latter since $km<1+km$), the hypothesis of Theorem 7.15 holds, and $\varepsilon(1+km)<km^2$ rearranges to $\frac{\varepsilon}{m(m-\varepsilon)}<k$, so $D(u_\tau)\le 1+\frac{\varepsilon}{m(m-\varepsilon)}<1+k=\frac{1}{4\sqrt{3}-6}$; Proposition 2.2(ii) converts this into the covering bound. For τ^* the medians ($m=\frac{1}{2}$) the box radius is $\frac{k}{4+2k}>\frac{1}{60}$ (equivalent to $58k>4$, i.e. to $10092>9801$), so Corollary 7.17 is strictly inside this region. □

Remark 7.19. The set of triples covered by Corollary 7.17 is a full-dimensional neighbourhood in the three-parameter space of cyclic triples, and the concurrent ones form a codimension-one subset of it: the corollary is a bound strictly inside non-concurrent territory, where by Theorem 6.11 no witness measure exists and $\sum\mathrm{rw}\ge 1$ remains open. We stress the honest comparison: the bound $4\sqrt{3}-6$ of [4] is uniform over *all* direction triples, while ours improves on it only on this neighbourhood; far from the concurrent locus the moment bound (Proposition 7.2) shows the transport method degrades irreparably — at the facets down to $\frac{2}{3}$.

Remark 7.20 (the moment bound is not exact: three certificates). The two sides of Theorem 7.7 sandwich $D(u)$ between finite, exactly verifiable computations: skeleton linear programs give certified *upper* bounds (Remark 7.14), explicit admissible ψ give certified *lower* bounds — and any feasible ψ, however found, is a rigorous bound. Piecewise-linear ψ with several knots do strictly beat the moment bound. Three archival certificates, each verified from scratch by exact rational arithmetic in the ancillary file `dual_certificates.py`:

triple	moment bound $\frac{1}{1-2\delta}$	certified dual bound
tilted facets, $\varepsilon = \frac{1}{10}$	$\frac{15}{11} \approx 1.3636$	$D \geq \frac{18}{13} \approx 1.3846$
two shared full edges, $(\frac{2}{3}, \frac{5}{6}, \frac{1}{12})$	$\frac{153}{142} \approx 1.0775$	$D \geq \frac{32}{29} \approx 1.1034$
cyclic sandwich, $\tau = (\frac{13}{25}, \frac{1}{2}, \frac{1}{2})$	$\frac{225}{224}$	$D > \frac{225}{224}$ (strict)

(Here *tilted facets* is the cyclic-symmetric triple with vertex values $(1, \varepsilon, 0)$, $(0, 1, \varepsilon)$, $(\varepsilon, 0, 1)$, and the second row is a triple with two shared full edges.) In particular $D(u) = (1 - 2\delta(u))^{-1}$ is *false* in general, although it holds at the facet triple in every dimension (Theorem 8.2) and trivially on the concurrent locus. For the sandwich parameter the certificates now give the strict two-sided bracket

$$\tfrac{225}{224} \ < \ D(u_\tau) \ \leq \ \tfrac{112}{111},$$

the upper bound being the optimal piecewise-uniform edge-skeleton measure; whether the upper bound is exact — equivalently, by Proposition 7.13 and Remark 7.14, whether some optimal measure is skeletal here — remains open.

The defect also behaves continuously enough to be mapped. Two statements, one general and one along a degeneration path:

Proposition 7.21 (lower semicontinuity)**.** *On families of n directions with fixed combinatorics, parametrized by their vertex values, D_K is lower semicontinuous.*

Proof. For fixed admissible ψ, the map $u \mapsto \min_{x \in K} \sum_i \psi_i(u_i(x))$ is continuous (the integrand is jointly continuous and K is compact). By Theorem 7.7, D_K is the supremum of these functions, hence lower semicontinuous. □

Remark 7.22 (a parallel degeneration path). Upper semicontinuity can fail to be obvious where the family degenerates — e.g. as two directions become parallel, where Theorem 3.1 (via [5]) makes the limiting pair have $D = 1$. Along the explicit path $u(\varepsilon)$ with vertex values $(0, 1, \frac{1}{3})$, $(0, 1, \frac{1}{3} + \varepsilon)$, $(\frac{1}{2}, 0, 1)$ (two shared full edges; $u_2 \to u_1$ as $\varepsilon \downarrow 0$), exact two-sided bounds hold for all $\varepsilon \in (0, \frac{2}{3})$:

$$1 + \frac{3\varepsilon}{2(3\varepsilon + 4)} \ \leq \ D\big(u(\varepsilon)\big) \ \leq \ 1 + \tfrac{3}{4}\varepsilon,$$

the lower bound from Theorem 7.11 (the dependency is $c = \big(\frac{\varepsilon + 4/3}{\varepsilon}, -\frac{4}{3\varepsilon}, 2\big)$, $c_0 = 1$), the upper from the explicit skeleton measure with edge masses $\big(\frac{1+3\varepsilon}{4}, \frac{2-3\varepsilon}{4}, \frac{1}{4}\big)$, whose marginal densities are all $\leq 1 + \frac{3}{4}\varepsilon$ (a symbolic verification is archived). So $D \to 1$ linearly at this parallel degeneration: no jump.

8. Higher-dimensional simplices: witness families and the facet defect

The transport-defect frame of §2 is dimension-free; this section records what we can prove beyond the plane. It is an extension of the main triangle results, not part of them. On Δ^d the uniform measure gives $D \leq d$ (Theorem 2.5) and the concurrence condition (Proposition 6.1) is necessary for $D = 1$ in every dimension. Two exact statements are within reach. First, witness families ($D = 1$) do exist beyond the plane: Theorem 8.1 constructs an explicit one-parameter family of concurrent quadruples on Δ^3 with skeleton witnesses. Second, the natural non-concurrent family — the facets — has its defect computed exactly in every dimension: $D_{\Delta^d}(\text{facets}) = \frac{d+1}{2}$ (Theorem 8.2), with a matched primal–dual certificate pair as in §7. Finally, Proposition 8.4 shows that for $d \geq 4$ the cyclic medians admit no 1-skeleton witness: supports must leave the skeleton.

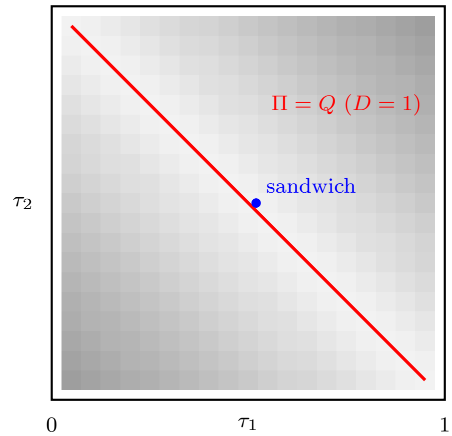

FIGURE 5. [Evidence, not proof.] The certified lower bound $(1-2\delta)^{-1}$ on $D(u_\tau)$ over the slice $\tau_3 = \frac{1}{2}$ of the cyclic family (closed form via Theorem 7.11 and Theorem 6.13; darker = larger, from 1 (white) to $\frac{3}{2}$ (black)). The concurrent locus in this slice is the anti-diagonal $\tau_1 + \tau_2 = 1$ (i.e. $\Pi = Q$), where $D = 1$; the defect grows toward the facet-like corners. Certified skeleton upper bounds match this map to within 0.06 on the strip $|\tau_1 + \tau_2 - 1| \leq \frac{1}{5}$ but degrade toward the corners (gap up to ≈ 0.59 at $(\frac{1}{20}, \frac{1}{20})$), where the optimal support leaves the boundary (Remark 7.14): the map is orientation for Problems 10.1 and 10.12, not a computation of D.

Let $\Delta^3 = \mathrm{conv}(e_1, \ldots, e_4)$ and fix $\sigma \in (0,1)$. Define four normalized affine directions by their vertex values (indices mod 4):

$$u_i(e_i) = 0, \qquad u_i(e_{i+1}) = 1, \qquad u_i(e_{i+2}) = \sigma, \qquad u_i(e_{i+3}) = 1 - \sigma. \tag{3}$$

Each row sums to 2, so the mid-hyperplanes $\{u_i = \frac{1}{2}\}$ concur at the centroid (Proposition 6.1); at $\sigma = \frac{1}{2}$ the $\{u_i = \frac{1}{2}\}$ are the "median" planes through an edge and the midpoint of the opposite edge. The four directions are pairwise non-parallel for every σ: an affine relation $u_{i+1} = c\,u_i + e$ forces, comparing vertex values, $(1-\sigma)^2 = 1$, and $u_{i+2} = c\,u_i + e$ forces $2\sigma(1-\sigma) = 0$, both impossible on $(0,1)$.

Theorem 8.1. *Let $\sigma \in (0, \frac{1}{2}]$ and set*

$$a(\sigma) = \frac{(1-\sigma)(1-2\sigma)}{2(\sigma^2 - 4\sigma + 2)}, \qquad b(\sigma) = \frac{\sigma(2-3\sigma)}{2(\sigma^2 - 4\sigma + 2)}.$$

The measure μ_σ placing mass $a(\sigma)$ on each of the four cycle edges e_ie_{i+1} and mass $b(\sigma)$ on each of the two diagonals e_1e_3, e_2e_4, uniformly in the parameter on each edge, is a probability measure on the 1-skeleton of Δ^3 with

$$u_{i\#}\mu_\sigma = \mathrm{Leb}[0,1] \qquad \text{for } i = 1, \ldots, 4.$$

Consequently every covering of Δ^3 by planks parallel to these four directions satisfies $\sum \mathrm{rw} \geq 1$. For $\sigma \in (\frac{1}{2}, 1)$ the same holds with the flip $u \mapsto 1-u$ (which exchanges $\sigma \leftrightarrow 1-\sigma$ and reverses the cyclic order). Moreover μ_σ is the unique cyclically invariant edge-uniform witness, and at $\sigma = \frac{1}{2}$ (where $a = 0$, $b = \frac{1}{2}$) it is the unique witness supported on the 1-skeleton at all.

Proof. Positivity of the weights on $(0, \frac{1}{2}]$: the roots of $\sigma^2 - 4\sigma + 2$ are $2 \pm \sqrt{2}$, so the denominator is positive on $(0, \frac{1}{2}] \subset (0, 2 - \sqrt{2})$, and the numerators are nonnegative there (vanishing only at $\sigma = \frac{1}{2}$, where the middle piece below is empty).

By the cyclic symmetry of (3) it suffices to compute the u_1-marginal. The six edges push forward as follows ($\sigma < \frac{1}{2}$; each edge uniform, mass as assigned): e_1e_2, values $0 \to 1$, density a on $[0, 1]$; e_2e_3, values $1 \to \sigma$, density $a/(1-\sigma)$ on $[\sigma, 1]$; e_3e_4, values $\sigma \to 1-\sigma$, density $a/(1-2\sigma)$ on $[\sigma, 1 - \sigma]$; e_4e_1, values $1 - \sigma \to 0$, density $a/(1 - \sigma)$ on $[0, 1 - \sigma]$; e_1e_3, values $0 \to \sigma$, density b/σ on $[0, \sigma]$; e_2e_4, values $1 \to 1 - \sigma$, density b/σ on $[1 - \sigma, 1]$. The total density is

$$a + \frac{a}{1-\sigma} + \frac{b}{\sigma} \ \text{ on } [0, \sigma] \text{ and on } [1 - \sigma, 1], \qquad a + \frac{2a}{1-\sigma} + \frac{a}{1-2\sigma} \ \text{ on } [\sigma, 1 - \sigma],$$

and a direct expansion shows both expressions equal 1 exactly for the stated $a(\sigma), b(\sigma)$; the total mass $4a + 2b = 1$ is then automatic, since pushforward preserves mass. The covering bound follows as in Proposition 2.2(ii) with $D = 1$. Uniqueness among cyclically invariant edge-uniform measures: the middle-piece equation determines a and either outer equation then determines b. (Exact linear algebra, recorded in the ancillary files, shows uniqueness among *all* edge-uniform weightings.) At $\sigma = \frac{1}{2}$ the values (3) make u_i *constant* ($\equiv \frac{1}{2}$) on the cycle edge $e_{i+2}e_{i+3}$; any skeleton measure giving positive mass to a cycle edge would push forward to an atom at $\frac{1}{2}$ under the corresponding direction, so all four cycle-edge masses vanish, and on the two diagonals each direction is affine onto $[0, \frac{1}{2}]$ or $[\frac{1}{2}, 1]$, forcing the uniform density $\frac{1}{2} + \frac{1}{2}$ split. □

Theorem 8.2 (facet defect in all dimensions)**.** *Let $d \geq 2$ and let $\lambda_1, \ldots, \lambda_{d+1}$ be the facet directions (barycentric coordinates) of Δ^d. Then*

$$D_{\Delta^d}(\lambda_1, \ldots, \lambda_{d+1}) \ = \ \frac{d+1}{2}.$$

The infimum is attained by the uniform measure on the closed polygonal cycle through the $d+1$ points

$$P_k = s^k\Big(0, \tfrac{1}{m}, \tfrac{2}{m}, \ldots, \tfrac{d}{m}\Big), \qquad k = 0, \ldots, d, \qquad m = \tfrac{d(d+1)}{2},$$

where s shifts the barycentric coordinates cyclically; all $d+1$ marginals equal $\frac{d+1}{2} \cdot \mathbf{1}_{[0,2/(d+1)]}$. The matching dual certificate is $\psi_i(t) = \big(1 - \frac{d+1}{2}\,t\big)_+$ for all i. For $d = 2$ the cycle is the inscribed triangle of Theorem 7.5.

Proof. Lower bound (dual certificate). Each $\psi_i(t) = (1 - \frac{d+1}{2}t)_+$ has integral $\frac{1}{d+1}$, so the tuple is admissible; pointwise $\psi_i(t) \geq 1 - \frac{d+1}{2}t$, so on Δ^d

$$\textstyle\sum_i \psi_i(\lambda_i) \ \geq \ (d+1) - \frac{d+1}{2}\sum_i \lambda_i \ = \ \frac{d+1}{2},$$

and Theorem 7.7 gives $D \geq \frac{d+1}{2}$. (Alternatively: $\sum_i \lambda_i = 1$ forces $\min_i \lambda_i(p) \leq \frac{1}{d+1}$ at every p, so $\delta = \frac{1}{2} - \frac{1}{d+1} = \frac{d-1}{2(d+1)}$, attained at the centroid, and Proposition 7.2 gives the same bound: the moment bound is exact here in every dimension.)

Upper bound (the cycle). The entries of each P_k are nonnegative and sum to $\big(0 + 1 + \cdots + d\big)/m = 1$, so the cycle lies in Δ^d; note $\frac{d}{m} = \frac{2}{d+1}$. Give each of the $d+1$ segments P_kP_{k+1} (indices mod $d+1$) mass $\frac{1}{d+1}$, uniformly in the parameter. Fix a coordinate j; its values at $P_0, P_1, \ldots, P_d$ are the cyclic rotation $\big(0, \frac{d}{m}, \frac{d-1}{m}, \ldots, \frac{1}{m}\big)$ of the base sequence, so along the cycle the coordinate traverses one rising segment $0 \to \frac{d}{m}$ and d falling segments, each dropping $\frac{1}{m}$. A uniform segment mass w whose values sweep an interval of length L pushes forward to density w/L on it. The rising segment contributes density $\frac{1/(d+1)}{d/m} = \frac{1}{2}$ on $[0, \frac{2}{d+1}]$; each falling segment contributes density $\frac{1/(d+1)}{1/m} = \frac{d}{2}$ on its value range $[\frac{r-1}{m}, \frac{r}{m}]$, and these d ranges tile $[0, \frac{d}{m}]$ exactly once. Total density: $\frac{1}{2}+\frac{d}{2}=\frac{d+1}{2}$ on $[0,\frac{2}{d+1}]$, i.e. the marginal is $\frac{d+1}{2}\mathbf{1}_{[0,2/(d+1)]}$ (a probability measure, as it must be). By the cyclic symmetry of the construction the same holds for every coordinate, so the cycle measure is $\frac{d+1}{2}$-admissible. □

Remark 8.3. The value is structural: it measures the exact reach of one-measure transport against the facet directions in every dimension, extending Remark 7.6 — transport certifies exactly $\sum \mathrm{rw} \geq \frac{2}{d+1}$ there, while the tiling Theorem 3.3 already gives the sharp $\sum \mathrm{rw} \geq 1$ for facet-parallel coverings. Note also that the optimal support is a one-dimensional loop through the *interior*: measures need not live on the boundary complex, a point made forcefully by the next proposition.

Proposition 8.4. *For $d \geq 4$, consider the cyclic medians of Δ^d: $u_i(e_i)=0$, $u_i(e_{i+1})=1$ and $u_i(e_j)=\frac{1}{2}$ otherwise (indices mod $d+1$). Then* every *edge e_je_k of Δ^d is constant for some direction, and consequently no measure supported on the* 1*-skeleton has uniform marginals for all $d+1$ directions.*

Proof. Fix an edge $\{j,k\}$. Removing j,k from the cycle on $d+1 \geq 5$ vertices leaves at least three vertices in at most two arcs, so some arc contains two cyclically consecutive vertices $\{m,m+1\}$ disjoint from $\{j,k\}$; then u_m takes the value $\frac{1}{2}$ at both e_j and e_k, hence is constant on the edge. A skeleton measure with positive mass on e_je_k pushes forward under u_m to a measure with an atom at $\frac{1}{2}$, which is not $\leq D\cdot\mathrm{Leb}$ for any finite D; so all edge masses vanish. □

Remark 8.5. The obstruction is specific to the all-$\frac{1}{2}$ interior values; whether distinct interior values (the $d \geq 4$ analogue of (3)) revive the skeleton, or the mass must move to 2-faces, is open and part of Problem 10.11.

9. A normalizer obstruction

For a direction u let $\ell_K(u)$ be the length of the longest chord of K parallel to u; always $\ell_K(u) \leq \mathrm{w}_K(u)$. Bakaev–Yehudayoff's chord bound $\sum_i \mathrm{w}(P_i)/\ell_K(u_i) \geq 1$ [4] follows from Bang's sign argument via the chord lemma. Consider interpolating the normalizer, $N_c=(1-c)\ell+c\,\mathrm{w}$, $c\in[0,1]$: $N_c=\ell$ at $c=0$ (chord bound) and $N_c=\mathrm{w}$ at $c=1$ (Conjecture 1.1).

Proposition 9.1. *For a convex body K, a direction u with $\|u\|=a/2$ and $\ell=\ell_K(u)\geq a$, the body $(K-u)\cap(K+u)$ contains a homothet of K of ratio $(\ell-a)/\ell$* [7, Lem. 2.3]*, and this ratio is sharp: for K a segment of length ℓ parallel to u, the set $(K-u)\cap(K+u)$ is a segment of length exactly $\ell-a$, and no homothet of K of larger ratio fits in it.*

Proof. The containment is [7, Lem. 2.3]. For sharpness let $K=[y,z]$ with $z-y=\ell\,\hat{u}$ and $u=\frac{a}{2}\hat{u}$: then $K-u=[y-\frac{a}{2}\hat{u},\, z-\frac{a}{2}\hat{u}]$ and $K+u=[y+\frac{a}{2}\hat{u},\, z+\frac{a}{2}\hat{u}]$, whose intersection is $[y+\frac{a}{2}\hat{u},\, z-\frac{a}{2}\hat{u}]$, a segment of length $\ell-a$; a homothet of K contained in it has ratio at most $(\ell-a)/\ell$. □

Remark 9.2 (methodological obstruction). The reading of Proposition 9.1 that matters here is about the *sign argument*: iterating the chord lemma over the directions u_j places a Bang-system witness $x_0+\sum_j \pm u_j$ as soon as $\sum_j \mathrm{w}(P_j)/\ell_j<1$, which is how the chord bound $\sum \mathrm{w}(P_j)/\ell_j \geq 1$ of [4] follows. The movement budget in direction u is governed by the homothety ratio of Proposition 9.1, which is exactly $(\ell-a)/\ell$ and cannot be stretched: the placement route proves the bound for the normalizer $N=\ell$ and for no direction-normalizer $N>\ell$; in particular it does not establish $\sum_i \mathrm{w}(P_i)/N_c(u_i)\geq 1$ for any $c>0$. This does *not* show the latter inequality false; it shows this route cannot prove it. Beating $2/(1+\sqrt{d})$ by such a normalizer therefore requires a witness that is not a Bang-system point, i.e. one exploiting the coupling of the covering — consistent with the sharpness of [4] for the cube.

10. Open problems and the covering-constant program

Research direction: the covering constants. The central open objects are $C_\Delta(u) \le C_\Delta^{111}(u)$ (Definition 2.3) for three pairwise non-parallel directions: Conjecture 1.1 restricted to such families says $C_\Delta(u) = 1$, and its one-plank-per-direction form is $C_\Delta^{111}(u) = 1$. What this paper establishes ($1/D \le C_\Delta \le C_\Delta^{111} \le 1$ throughout):

family	$1/D_\Delta$	C_Δ	C_Δ^{111}
two directions	1	1	1
concurrent cyclic (incl. medians)	1	1 (rigid)	1
all sharing one full edge	1	1	1
facet triple	2/3	1 (tiling)	1
cyclic, $\max_i \lvert\tau_i - \frac{1}{2}\rvert \le \frac{1}{60}$	$\ge 29/31$	$[29/31, 1]$	$[29/31, 1]$
sandwich $(\frac{13}{25}, \frac{1}{2}, \frac{1}{2})$	$[\frac{111}{112}, \frac{224}{225})$	$[\frac{111}{112}, 1]$	1 (Thm 10.8)
two shared full edges, $(\frac{2}{3}, \frac{5}{6}, \frac{1}{12})$	$[\frac{60}{71}, \frac{29}{32}]$	$[\frac{60}{71}, 1]$	open

Problem 10.1. Is $C_\Delta(u) = 1$ (equivalently, is the gap $G_{\mathrm{tr}} = C_\Delta - 1/D$ closed to the covering truth) for every pairwise non-parallel triple on the triangle? And is $C_\Delta^{111}(u) = 1$, its one-plank-per-direction form? Theorem 6.11 settles the concurrent case (transport, sharp and rigid) and Theorem 3.3 the facet triple (tiling, any number of planks); Theorem 10.8 settles $C_\Delta^{111} = 1$ at one tilted non-concurrent triple. For non-concurrent non-facet triples in general both remain open — the three-plank sub-problem in quantitative form. (Conjecture 1.1 for the triangle concerns arbitrarily many planks in arbitrary directions and is not reduced to it.)

Two structural remarks delimit the attack. First, the boundary is forced:

Lemma 10.2 (edge reduction)**.** *If planks P_1, P_2, P_3 (one per direction) cover Δ, then on each edge the traces cover the edge; consequently, on each edge with roles (c, f, g) as in Appendix A, $r_c/\tau_c + r_f + r_g/(1-\tau_g) \ge 1$. These necessary linear conditions alone do* not *suffice: their linear-programming value lies well below 1 (e.g. $\frac{3}{5}$ at the medians), so any proof of Problem 10.1 must couple the boundary constraints with an interior mechanism.*

Proof. A covering of Δ covers each edge, on which the trace of P_i is an interval of length at most $r_i/|\mathrm{slope}|$, with the slopes of Appendix A; the medians value is the symmetric LP $2r_c + r_f + 2r_g \ge 1$ on three cyclic constraints, minimized at $\sum r = \frac{3}{5}$. ☐

Second, an uncovered point lives in one of eight explicit *sign cells* $\{u_i < l_i$ or $u_i > h_i\}$, and the machinery of Lemma 6.8 — the affine dependency $\sum c_i u_i \equiv c_0$, which exists for *every* triple (Proposition 7.9(i)), and the Pocket Lemma — controls those cells for arbitrary intervals (l_i, h_i): this is how Theorem 6.13 produces canonical coverings for all cyclic triples. The step that fails for $\sum r < 1$ is that the boundary is no longer *tiled*, only covered; on the cyclic family the case split is (a) some edge uncovered — immediate contradiction by Lemma 10.2 — and (b) all edges covered with overlaps, where a hybrid certificate (a measure argument on the boundary combined with a sumset/tiling obstruction as in Theorem 3.3) appears to be needed; formalizing it is the program, not a result of this paper. Exact experiments (archived in `covering_constant.py`) support the conjecture: float-guided minimization of $\sum r$ over exactly-verified coverings of the sandwich and two further non-concurrent triples never produced a certified covering with $\sum r < 1$.

A partially computer-assisted case of the program. We determine C_Δ^{111} at one non-concurrent triple: the *sandwich* $\tau_0 = (\frac{13}{25}, \frac{1}{2}, \frac{1}{2})$ of Remark 7.20 ($\Pi = \frac{13}{100} \ne \frac{12}{100} = Q$, so it is non-concurrent by Theorem 6.13). Both the facet triple (Theorem 3.3, tiling) and the concurrent triples (Theorem 6.11, transport) are already settled; the sandwich is, to our knowledge, the first *genuinely tilted* case — non-facet and non-concurrent — established, and it is settled only for one plank per direction (C^{111}), the full covering constant $C_\Delta(\tau_0)$ remaining open. The proof is by hand except for one finite *certificate* check, isolated below and discharged by an exact-rational computation verified by an independent checker (Appendix B); we mark the boundary explicitly.

Throughout, a covering is by three planks $P_i = \{u_i \in [l_i, h_i]\}$, one per direction, $r_i = h_i - l_i \geq 0$; recall $C^{111}_\Delta(\tau_0)$ is the infimum of $\sum_i r_i$ over such coverings (Definition 2.3). The first tool is the two-direction case as a local instrument.

Lemma 10.3 (two-plank tool)**.** *Let $K' \subseteq \Delta$ be a convex body and u_j, u_k two non-parallel directions with extents $w_j = |u_j(K')|$, $w_k = |u_k(K')| > 0$. If two planks $\{u_j \in I_j\}$, $\{u_k \in I_k\}$ cover K', then $|I_j|/w_j + |I_k|/w_k \geq 1$.*

Proof. Gardner's theorem [5, Theorem 1] applied to the convex body K' and the directions u_j, u_k gives a probability measure ν on K' whose pushforward under each of u_j, u_k is uniform on the respective range; so $\nu(\{u_j \in I_j\}) \leq |I_j|/w_j$ and likewise for k. Covering gives $1 = \nu(K') \leq |I_j|/w_j + |I_k|/w_k$. $\square$

Theorem 10.4 (extreme regime)**.** *For a cyclic parameter $\tau \in (0,1)^3$ put*

$$w_0(\tau) = \tfrac{1}{2} \min_{\text{cyclic } \varrho} \min\Big(\tau_1,\, 1-\tau_1,\, \tfrac{(1-\tau_2)\tau_1}{1+\tau_1},\, \tfrac{\tau_3(1-\tau_1)}{2-\tau_1}\Big),$$

the inner minimum taken with (τ_1, τ_2, τ_3) cyclically rotated; then $w_0(\tau) > 0$ and $w_0(\tau_0) = \frac{2}{25}$. If a covering of Δ by three planks in the directions of τ has $\max_i r_i \geq 1 - w_0(\tau)$, then $\sum_i r_i \geq 1$, with equality only for a trivial covering (some $r_i = 1$, the others 0).

Proof. By the cyclic symmetry it suffices to treat the case where P_1 is the wide plank; write $l = l_1$, $s = 1 - h_1$, so $r_1 = 1 - l - s$ and $l + s \leq w_0$. The part of Δ not covered by P_1 is $K_{\text{lo}} = \{u_1 \leq l\}$ (a triangle at the vertex B, empty iff $l = 0$) and $K_{\text{hi}} = \{u_1 \geq 1 - s\}$ (a triangle at C, empty iff $s = 0$); both are genuine corner triangles since $l, s < \min(\tau_1, 1-\tau_1)$. Since each u_i is affine, its range over a corner triangle is the interval between its values at the three corner vertices; a direct computation (verified symbolically in `c3_extreme_regime.py`) gives the ranges and extents

$$u_2(K_{\text{lo}}) = [1 - \tfrac{l}{\tau_1}, 1],\ a_2 = \tfrac{l}{\tau_1};\quad u_3(K_{\text{lo}}) = [\tau_3(1-l), \tau_3 + \tfrac{(1-\tau_3)l}{\tau_1}],\ a_3 = \tfrac{l\,g_2}{\tau_1};$$
$$u_2(K_{\text{hi}}) = [\tau_2(1 - \tfrac{s}{1-\tau_1}), \tau_2 + (1-\tau_2)s],\ b_2 = \tfrac{s\,g_3}{1-\tau_1};\quad u_3(K_{\text{hi}}) = [0, \tfrac{s}{1-\tau_1}],\ b_3 = \tfrac{s}{1-\tau_1},$$

with $g_2 = 1 - \tau_3(1-\tau_1)$, $g_3 = 1 - \tau_1(1-\tau_2)$ the dependency factors of Theorem 6.13. All four extents strictly exceed the corner's u_1-width: $a_2 - l = \frac{l(1-\tau_1)}{\tau_1}$, $a_3 - l = \frac{l(1-\tau_1)(1-\tau_3)}{\tau_1}$, $b_2 - s = \frac{s\tau_1\tau_2}{1-\tau_1}$, $b_3 - s = \frac{s\tau_1}{1-\tau_1}$, all > 0.

Case A: no plank meets both corners. Each nonempty corner is covered by the planks touching it. If $P_2 \supseteq K_{\text{lo}}$ and $P_3 \supseteq K_{\text{hi}}$ then $r_2 \geq a_2 > l$ and $r_3 \geq b_3 > s$; if instead $P_2 \supseteq K_{\text{hi}}$, $P_3 \supseteq K_{\text{lo}}$ then $r_2 \geq b_2 > s$ and $r_3 \geq a_3 > l$. If both P_2, P_3 are needed on one corner (forcing the other empty, so $s = 0$ or $l = 0$), the two-plank tool (Lemma 10.3) on that corner gives $r_2 + r_3 \geq \min$ of its two extents $\geq l$ (resp. $\geq s$). In every sub-case $\sum r = (1 - l - s) + (r_2 + r_3) \geq 1$, strict unless $l = s = 0$.

Case B: some plank meets both corners. Its interval spans the gap between the two disjoint ranges. If P_2 meets both, $r_2 \geq (1 - \frac{l}{\tau_1}) - (\tau_2 + (1-\tau_2)s)$, which is $\geq l + s$ because $w_0 \leq \frac{(1-\tau_2)\tau_1}{2(1+\tau_1)}$; if P_3 meets both, $r_3 \geq \tau_3(1-l) - \frac{s}{1-\tau_1} \geq l + s$ because $w_0 \leq \frac{\tau_3(1-\tau_1)}{2(2-\tau_1)}$. Either way $\sum r \geq (1 - l - s) + (l + s) = 1$, strict unless $l = s = 0$. Equality thus forces $l = s = 0$, i.e. $r_1 = 1$: a trivial covering. $\square$

The extreme regime leaves the *balanced* coverings ($r_i < 1 - w_0$ for all i). We reduce these to a finite computation. Write a covering as a point $x = (l_1, h_1, l_2, h_2, l_3, h_3) \in [0,1]^6 =: \mathcal{C}$, call it *proper* if all $r_i > 0$; bisection produces axis-aligned boxes $B \subseteq \mathcal{C}$ with coordinate ranges $l_i \in [l_i^-, l_i^+]$, $h_i \in [h_i^-, h_i^+]$. Define four *pruning predicates* on a box B:

(P1) $\sum_i \max(0, h_i^- - l_i^+) > 1$;
(P2) $h_i^- - l_i^+ \geq 1 - w_0$ for some i;
(P4) $l_i^- > h_i^+$ for some i;
(P3) the enlarged configuration $I_i^+ = [l_i^-, h_i^+]$ does not cover Δ.

Lemma 10.5 (soundness of the prunes)**.** *If any of* (P1)–(P4) *holds for B, then no proper covering in B has $\sum_i r_i \leq 1$. Indeed* (P1) *gives $\sum r_i \geq \sum \max(0, h_i^- - l_i^+) > 1$ pointwise;* (P2) *gives $r_i \geq 1 - w_0$, so Theorem 10.4 applies;* (P4) *forces plank i empty, so at most two non-parallel directions are used and Theorem 3.1 gives $\sum r \geq 1$;* (P3) *gives $[l_i, h_i] \subseteq I_i^+$ for every configuration in B, so $\bigcup_i P_i \subseteq \bigcup_i \{u_i \in I_i^+\}$ and a point uncovered by the latter is uncovered by the former — no configuration in B covers.*

Proof. Each clause is the displayed monotonicity in $l_i^\pm, h_i^\pm$; (P2), (P4) invoke the two named theorems. □

Lemma 10.6 (complete non-covering test)**.** *A finite union $\bigcup_i \{u_i \in I_i^+\}$ fails to cover Δ if and only if either* (a) *some edge of Δ is not covered by the traces of the I_i^+, or* (b) *some sign cell $\{u_i < l_i^+ \text{ or } u_i > h_i^+\}$ has positive area in Δ. Both tests are exact and rational.*

Proof. An uncovered point lies in exactly one sign cell (a relatively open convex polygon). If interior to Δ its cell has positive area, case (b); if on $\partial\Delta$ it lies on an edge whose traces miss it, case (a). Conversely each condition exhibits an uncovered point. Areas and endpoints are rational in τ, l^+, h^+. □

The reduction is a *static* statement about a given finite tree — not a claim that any search terminates.

Lemma 10.7 (finite covering certificate)**.** *Let $\mathcal{T}$ be a finite binary tree of axis-aligned boxes with root $\mathcal{C} = [0,1]^6$, in which each internal node's box is split by one coordinate into the boxes of its two children, and each leaf box satisfies one of* (P1)–(P4)*. Then no proper balanced covering of Δ with $\sum_i r_i \leq 1$ exists; indeed every proper balanced covering has $\sum_i r_i > 1$.*

Proof. By induction on the tree, using the bisection property at each internal node, the leaf boxes cover the root $\mathcal{C}$. A proper balanced covering with $\sum r \leq 1$ would be a point of $\mathcal{C}$, hence of some leaf box B; but B satisfies one of (P1)–(P4), which by Lemma 10.5 excludes any proper covering in B with $\sum r \leq 1$ — a contradiction. Strictness: such a covering is proper (rules out (P4)) and covers (rules out (P3)); if it were balanced its leaf could not be (P2); so it is (P1), whose closed box has $\min \sum r > 1$, whence $\sum r > 1$. □

Theorem 10.8 ($C_\Delta^{111}(\tau_0) = 1$; *partially computer-assisted*)**.** *For the sandwich $\tau_0 = (\frac{13}{25}, \frac{1}{2}, \frac{1}{2})$, every covering of Δ by three planks, exactly one parallel to each direction, satisfies $\sum_i \mathrm{rw} \geq 1$; the infimum is attained (by a single full plank), so $C_\Delta^{111}(\tau_0) = 1$. This is the first genuinely tilted — non-facet, non-concurrent — triple on the triangle for which the three-plank bound is established, to our knowledge.*

Proof. Partition three-plank coverings into (i) extreme ($\max r_i \geq 1 - w_0$), (ii) some plank empty, and (iii) proper balanced. Case (i) is Theorem 10.4 ($\sum r \geq 1$); case (ii) is Theorem 3.1 ($\sum r \geq 1$, at most two non-parallel directions). For case (iii), by Lemma 10.7 it suffices to exhibit a finite binary partition tree of $\mathcal{C}$ with root $\mathcal{C}$, coordinate bisections at internal nodes, and every leaf satisfying one of (P1)–(P4): then every proper balanced covering has $\sum r > 1$. *Such a tree is provided as an explicit certificate and verified in exact rational arithmetic by an independent checker* (sharing no code with any search) that reconstructs every box, confirms the bisection structure and that the leaves cover $\mathcal{C}$, and re-validates every leaf's predicate (Appendix B); the tree has 812,651 leaves. The infimum 1 is attained by $I_1 = [0,1]$, $I_2 = I_3 = \varnothing$. $\square$

Remark 10.9 (the human/machine boundary). Every step above is proved by hand except the existence of the finite certificate tree of Lemma 10.7 — a static, finite object whose validity (correct bisection structure; every leaf soundly pruned) is checked in *exact* rational arithmetic (no floating point, independent of hardware) but at a scale (812,651 leaves) beyond hand verification. This is what makes Theorem 10.8 partially computer-assisted; its validity rests on the independent *checker* and the *certificate* (Appendix B), *not* on any search — the search is merely how the certificate was found and is irrelevant to the proof. This settles C_Δ^{111} at one triple, not the family, and leaves $C_\Delta(\tau_0)$ (arbitrarily many planks) open; it does not bear on Conjecture 1.1 (Problem 10.1).

Remark 10.10 (equality; auxiliary). Equality $\sum \mathrm{rw} = 1$ holds for the trivial coverings above. No *proper* two-plank tight covering exists: an auxiliary exact optimization over the image triangle (secondary verification, `r15_equality.py`, not part of the certificate) gives minimum widths $\frac{31}{25}$ (pairs containing u_1) and $\frac{5}{4}$ ((u_2, u_3)), both > 1; with case (iii) strict, no nontrivial covering is tight. This characterization is not needed for $C_\Delta^{111}(\tau_0) = 1$.

Problem 10.11. Theorem 6.11 settles the existence of uniform-marginal witness measures for three directions on the triangle, and Theorem 8.1 gives an affirmative one-parameter family of concurrent quadruples on Δ^3. For $d \geq 3$ the concurrence condition of Proposition 6.1 remains necessary; is it sufficient for $d+1$ directions on Δ^d? For the cyclic medians of Δ^d with $d \geq 4$, any witness must vanish on the 1-skeleton (Proposition 8.4); the first open cases are supports on 2-faces, interior vertex values as in (3), or one-dimensional interior supports as in Theorem 8.2.

Problem 10.12. Is $\sup_u D(u) = \frac{3}{2}$ over pairwise non-parallel triples on the triangle, attained (only, by Proposition 7.4) at the facet triple? What is known: the supremum lies in $[\frac{3}{2}, 2]$ (Theorems 7.5 and 2.5); the moment bound never certifies more than $\frac{3}{2}$ ($\delta(u) \leq \frac{1}{6}$ for every family, Theorem 7.3), and it is *not* exact in general (Remark 7.20), so by Theorem 7.7 a triple with $D(u) > \frac{3}{2}$ would have to be exposed by a multi-knot ψ — exact searches near the facet-like degenerations have so far produced certified bounds approaching $\frac{3}{2}$ only from below. A positive answer would give the universal bound $\sum \mathrm{rw} \geq \frac{2}{3}$ for coverings of the triangle by planks parallel to *any* three directions — well below $4\sqrt{3} - 6$, so of structural rather than quantitative interest. Related: characterize where the skeleton computes the defect ($D_\partial = D$; Remark 7.14) — the support's phase transition visible in Figure 5.

Problem 10.13. Whether $\sum \mathrm{w}(P_i)/N_c(u_i) \geq 1$ holds for some $c > 0$ (a coupling certificate), which would improve the general bound past $2/(1 + \sqrt{d})$.

Appendix A. Edge tilings of cyclic triples

Throughout, $u_1 = (\tau_1, 0, 1)$, $u_2 = (0, 1, \tau_2)$, $u_3 = (1, \tau_3, 0)$ is a cyclic triple with arbitrary $\tau_1, \tau_2, \tau_3 \in (0,1)$ (*no* concurrence is assumed), and $I_i = [l_i, h_i] \subset [0,1]$, $l_i < h_i$, are three intervals. Say (I_1, I_2, I_3) satisfies the *edge-tiling conditions* if on each edge of Δ the traces $T_i = \{x \in \text{edge} : u_i(x) \in I_i\}$ cover the edge and have pairwise disjoint interiors. Indices are mod 3.

Proposition A.1. *For every $\tau \in (0,1)^3$ the edge-tiling conditions have exactly three solutions: with*

$$\lambda_i = \frac{\tau_i\big(1-\tau_{i+1}(1-\tau_{i+2})\big)}{1+\tau_1\tau_2\tau_3}, \qquad \rho_i = \frac{1-\tau_{i+2}(1-\tau_i)}{1+(1-\tau_1)(1-\tau_2)(1-\tau_3)},$$

they are $\big([\lambda_i,\rho_i]\big)_i$, $\big([0,\lambda_i]\big)_i$, and $\big([\rho_i,1]\big)_i$. One has $0<\lambda_i<\tau_i<\rho_i<1$ for all i and all τ. For $\tau_i \equiv \frac{1}{2}$ (the medians) the solutions are $[\frac{1}{3},\frac{2}{3}]^3$, $[0,\frac{1}{3}]^3$, $[\frac{2}{3},1]^3$.

Setup. Parametrize the edges by $t \in [0,1]$ in the orientations $A \to B$, $B \to C$, $C \to A$. From the vertex values, each edge sees one direction with values $[0,\tau_c]$ (the *low* direction c, $u_c(t) = \tau_c(1-t)$), one with values $[0,1]$ (the *full* direction f, $u_f(t)=t$), and one with values $[\tau_g,1]$ (the *high* direction g, $u_g(t) = 1-(1-\tau_g)t$), with roles

$$(c,f,g) = (1,2,3) \text{ on } AB, \qquad (3,1,2) \text{ on } BC, \qquad (2,3,1) \text{ on } CA.$$

Inverting, the traces on an edge with roles (c,f,g) are

$$\begin{aligned} T_c &= \Big[\max\big(0,1-\tfrac{h_c}{\tau_c}\big),\, 1-\tfrac{l_c}{\tau_c}\Big] \text{ (nondegenerate iff } l_c<\tau_c), \qquad T_f = [l_f,h_f], \\ T_g &= \Big[\tfrac{1-h_g}{1-\tau_g},\, \min\big(1,\tfrac{1-l_g}{1-\tau_g}\big)\Big] \text{ (nondegenerate iff } h_g>\tau_g). \end{aligned} \tag{4}$$

Classify each I_i by its position relative to τ_i: type L if $h_i \le \tau_i$, type R if $l_i \ge \tau_i$, type M if $l_i < \tau_i < h_i$; every I_i is of exactly one type. Two elementary observations, used throughout:

(O1) If I_c is of type M then $T_c = [0,\, 1-l_c/\tau_c]$ (it contains the left endpoint); if I_c is of type R then T_c is degenerate (a point or empty). Dually, if I_g is of type M then $T_g = [(1-h_g)/(1-\tau_g),\, 1]$; if I_g is of type L then T_g is degenerate.

(O2) The tiling condition is insensitive to degenerate traces, in both of its clauses. *Covering:* a degenerate trace covers at most one point, and since the nondegenerate traces are finitely many closed sets whose union contains $[0,1]$ minus finitely many points, that union is all of $[0,1]$. *Disjointness:* a degenerate trace has empty interior, so the disjoint-interiors condition involving it is vacuous — it obstructs nothing. Hence discarding degenerate traces loses no solution and creates none: the closed intervals (I_1,I_2,I_3) satisfy the edge-tiling conditions if and only if on each edge the *nondegenerate* traces tile $[0,1]$; in particular, finitely many closed intervals of positive length with disjoint interiors covering $[0,1]$ abut consecutively from 0 to 1. Consequently the case analysis below requires no genericity in τ or in the endpoints: whenever an endpoint coincidence (e.g. $h_i=\tau_i$ or $l_i=\tau_i$) makes a trace degenerate, (O2) discards it, and every case argument uses only the defining non-strict inequalities of the types.

Symmetries. Two operations act on the data $(\tau; I_1,I_2,I_3)$ and transport edge-tilings to edge-tilings.

The *rotation* ϱ is induced by the affine map $A \to B \to C \to A$ of Δ: it carries the cyclic triple with parameter (τ_1,τ_2,τ_3) to the one with parameter (τ_2,τ_3,τ_1) and acts by $(I_1,I_2,I_3) \mapsto (I_2,I_3,I_1)$, shifting type vectors cyclically.

The *reflected flip* ς is the affine reflection of Δ swapping B and C, composed with the flip $s \mapsto 1-s$ of each normalized direction. The flip replaces each u_i by $1-u_i$ and each I_i by $1-I_i$ $(1-[l,h]=[1-h,1-l])$, leaving every plank, every width, every trace and the tiling conditions unchanged; the reflection then restores the cyclic normalization (value 0 and 1 at the prescribed vertices), and the composite carries the parameter (τ_1,τ_2,τ_3) to $(1-\tau_1,\, 1-\tau_3,\, 1-\tau_2)$, acting by $(I_1,I_2,I_3) \mapsto (1-I_1,\, 1-I_3,\, 1-I_2)$; on type vectors it acts by $(X_1,X_2,X_3) \mapsto (\bar X_1, \bar X_3, \bar X_2)$, where the bar exchanges L $\leftrightarrow$ R and fixes M.

One checks $\varsigma^2 = \mathrm{id}$ and $\varsigma\varrho\varsigma = \varrho^{-1}$, so $G = \langle \varrho,\varsigma\rangle$ is a group of order 6 (isomorphic to S_3). Each $g \in G$ maps the edge-tiling problem with parameter τ *bijectively* onto the problem with parameter $g(\tau)$. Since Proposition A.1 is proved for *all* $\tau \in (0,1)^3$ simultaneously, it suffices to treat one representative of each G-orbit of the 27 type vectors $(X_1, X_2, X_3) \in \{\mathrm{L}, \mathrm{M}, \mathrm{R}\}^3$:

$$\mathrm{MMM}; \quad \mathrm{LLL}\ (\sim \mathrm{RRR}); \quad \mathrm{MML}; \quad \mathrm{LLM}; \quad \mathrm{LLR}; \quad \mathrm{LMR}; \quad \mathrm{LRM}$$

(orbit sizes $1, 2, 6, 6, 6, 3, 3$, summing to 27). The outcome of the case analysis below, at a glance — each impossible case with its labelled mechanism:

type	orbit	outcome	mechanism
MMM	1	solution ($[\lambda_i, \rho_i]$)	two decoupled cyclic systems, unique fixed points
LLL $\sim$ RRR	2	solutions ($[0, \lambda_i]$), ($[\rho_i, 1]$)	forced endpoints $l_c = 0$ ($h_g = 1$)
MML	6	impossible	forced full-line trace $\Rightarrow$ empty-interior contradiction
LLM	6	impossible	forced full-line trace $\Rightarrow$ empty-interior contradiction
LLR	6	impossible	lone trace must tile: endpoint $h_1 = 1$ contradicts type L
LMR	3	impossible	forced abutments $\Rightarrow$ interior overlap above l_2
LRM	3	impossible	$I_3 = [0, 1]$ forced $\Rightarrow$ empty-interior contradiction

Case MMM. By (O1), on every edge $T_c = [0,\, 1 - l_c/\tau_c]$ and $T_g = [(1-h_g)/(1-\tau_g),\, 1]$, both nondegenerate, and T_f has positive length. Disjointness of interiors forces T_f inside the gap between them, and covering forces the abutments

$$l_f = 1 - \frac{l_c}{\tau_c}, \qquad h_f = \frac{1-h_g}{1-\tau_g}.$$

Reading this on the three edges gives two decoupled cyclic systems,

$$l_2 = 1 - \frac{l_1}{\tau_1}, \quad l_1 = 1 - \frac{l_3}{\tau_3}, \quad l_3 = 1 - \frac{l_2}{\tau_2}; \qquad h_1 = \frac{1-h_2}{1-\tau_2}, \quad h_2 = \frac{1-h_3}{1-\tau_3}, \quad h_3 = \frac{1-h_1}{1-\tau_1}.$$

Each system composes three decreasing affine maps, so the composite self-map (of l_1, resp. h_1) is affine with negative slope ($-1/(\tau_1\tau_2\tau_3)$, resp. $-1/\prod(1-\tau_i)$), hence has a unique fixed point: the solutions are $l_i = \lambda_i$, $h_i = \rho_i$ (direct substitution verifies the formulas; e.g. $1 - \lambda_1/\tau_1 = \big(\tau_1\tau_2\tau_3 + \tau_2(1-\tau_3)\big)/(1+\tau_1\tau_2\tau_3) = \lambda_2$). The chain $0 < \lambda_i < \tau_i < \rho_i < 1$ reduces to $\tau_{i+1}(\tau_{i+2}-1) < \tau_1\tau_2\tau_3$ and $\tau_i(1-\tau_{i+1}) < 1$, both automatic; so the solution is of type MMM and, by construction, tiles each edge as $[0, l_f] \cup [l_f, h_f] \cup [h_f, 1]$ (Figure 6).

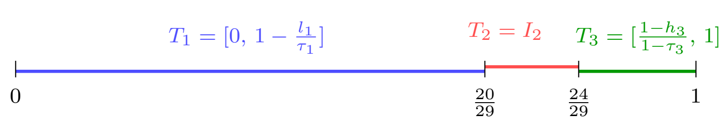

FIGURE 6. The MMM edge tiling on AB (roles $(c, f, g) = (1, 2, 3)$) for the running example $\tau = (\frac{5}{9}, \frac{4}{5}, \frac{1}{6})$: the traces of $I = ([\frac{5}{29}, \frac{25}{29}], [\frac{20}{29}, \frac{24}{29}], [\frac{4}{29}, \frac{9}{29}])$ abut at $\frac{20}{29}$ and $\frac{24}{29}$ and tile $[0, 1]$, as forced in Case MMM.

Case LLL. All T_g are degenerate, so each edge is tiled by $T_c = [1 - h_c/\tau_c,\, 1 - l_c/\tau_c]$ (the full image, as $I_c \subset [0, \tau_c]$) and $T_f = I_f$ with $h_f \le \tau_f < 1$. The tile containing 1 must be T_c, forcing $l_c = 0$. If $h_c = \tau_c$ then $T_c = [0, 1]$ and T_f has empty interior, impossible; so T_f contains 0 ($l_f = 0$) and abuts T_c: $h_f = 1 - h_c/\tau_c$. On the three edges: $l_i = 0$ for all i, and the h_i satisfy the *same* cyclic system as the l_i in case MMM, so $h_i = \lambda_i$, giving $\big([0, \lambda_i]\big)_i$ (type L since $\lambda_i < \tau_i$). Applying the reflection symmetry, the orbit representative RRR yields the unique solution $h_i = 1$, $l_i = \rho_i$, i.e. $\big([\rho_i, 1]\big)_i$; equivalently, the l_i satisfy the same cyclic system as the h_i in case MMM.

Case MML (impossible). On AB: T_3 is degenerate (I_3 of type L), so $[0, 1]$ is tiled by $T_1 = [0,\, 1 - l_1/\tau_1]$ (O1) and $T_2 = I_2$. If $1 \in T_1$ then $T_1 = [0, 1]$ and T_2 has empty interior, impossible; so $h_2 = 1$. Then on BC the high trace (4) is $T_2 = [\frac{1-h_2}{1-\tau_2}, \min(1, \frac{1-l_2}{1-\tau_2})] = [0, 1]$ (as $h_2 = 1$ and $l_2 < \tau_2$), leaving the full trace $T_1 = I_1$ (positive length) with empty interior — contradiction.

Case LLM (impossible). On BC ($(c, f, g) = (3, 1, 2)$): the high trace is degenerate (I_2 of type L), so $[0, 1]$ is tiled by $T_3 = [0,\, 1 - l_3/\tau_3]$ (O1, I_3 of type M) and $T_1 = I_1$ with $h_1 \le \tau_1 < 1$. The tile containing 1 must be T_3, so $l_3 = 0$ and $T_3 = [0, 1]$, leaving T_1 with empty interior — contradiction.

Case LLR (impossible). On BC both compressed traces are degenerate (I_3 of type R gives a degenerate low trace, I_2 of type L a degenerate high trace), so $T_1 = I_1$ alone tiles $[0, 1]$, forcing $h_1 = 1 > \tau_1$ — contradicting type L.

Case LMR (impossible). *On BC* ($(c, f, g) = (3, 1, 2)$): the low trace is degenerate (I_3 of type R), so $T_1 = I_1$ and $T_2 = [\frac{1-h_2}{1-\tau_2}, 1]$ (O1, I_2 of type M) tile $[0, 1]$. If $T_2 = [0, 1]$ then T_1 has empty interior, impossible; so T_1 contains 0 and abuts T_2: $l_1 = 0$ and $h_1 = \frac{1-h_2}{1-\tau_2}$.

On CA ($(c, f, g) = (2, 3, 1)$): the high trace is degenerate (I_1 of type L), so $T_2 = [0,\, 1 - l_2/\tau_2]$ (O1) and $T_3 = I_3$ tile $[0, 1]$. If $T_2 = [0, 1]$ then T_3 has empty interior, impossible; so $h_3 = 1$ and $l_3 = 1 - l_2/\tau_2$. Since I_3 has positive width, $l_3 < 1$, which forces $l_2 > 0$.

On AB ($(c, f, g) = (1, 2, 3)$): as $h_3 = 1$, the high trace is $T_3 = \big[0, \min\big(1, \frac{1-l_3}{1-\tau_3}\big)\big]$. Two subcases. If $\frac{1-l_3}{1-\tau_3} \ge 1$, then $T_3 = [0, 1]$ and the full trace $T_2 = I_2$ (positive length) has empty interior — contradiction. Otherwise $T_3 = \big[0, \frac{1-l_3}{1-\tau_3}\big] = \big[0, \frac{l_2}{\tau_2(1-\tau_3)}\big]$; since $\tau_2(1 - \tau_3) < 1$ and $l_2 > 0$, its right endpoint strictly exceeds l_2, so the interiors of T_3 and of $T_2 = [l_2, h_2]$ overlap on an interval just above l_2 — contradiction.

Case LRM (impossible). On CA: both compressed traces are degenerate (I_2 of type R, I_1 of type L), so $T_3 = I_3$ alone tiles $[0, 1]$, i.e. $I_3 = [0, 1]$. Then on AB the high trace (4) is $T_3 = [\frac{1-h_3}{1-\tau_3}, \min(1, \frac{1-l_3}{1-\tau_3})] = [0, 1]$, leaving $T_2 = I_2$ (positive length) with empty interior — contradiction.

All orbits are exhausted, proving Proposition A.1. □

The three solutions have also been verified by exact rational searches at $\tau = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ and $\tau = (\frac{1}{2}, \frac{2}{3}, \frac{1}{3})$, and the median enumeration was verified by two further independent exact computations; all scripts are archived as ancillary files (Appendix B).

Appendix B. Ancillary files and verification protocol

This paper is proved by hand throughout, *with a single exception*: Theorem 10.8 ($C^{111}_{\Delta}(\tau_0) = 1$ at the sandwich τ_0) reduces, by the human Lemmas 10.3–10.7 and Theorem 10.4, to the emptiness of an explicit finite bisection tree $T(\tau_0)$, discharged by an exact-rational computation and re-verified by the independent certificate checker described below; all other results are proved in full in the text or in Appendix A. Apart from that one lemma, the ancillary scripts are *secondary verification*: exact re-derivations — rational or symbolic arithmetic throughout, never floating point — of load-bearing identities and finite case enumerations, archived so that a reader can re-check them mechanically. Where a script enumerates cases (tiling types, role patterns), the enumeration is exhaustive over an explicitly finite index set, so no case is lost by sampling. For each script we record what it verifies and which statement it accompanies; all run under Python 3 with SymPy (or only integer/rational standard libraries where noted).

The certified computation (Theorem 10.8). Following the search-versus-verification separation, the emptiness of $T(\tau_0)$ is established in three artifacts. The *searcher* `c3_balanced_bb.py` grows $T(\tau_0)$ by depth-first bisection, pruning by (P1)–(P4); it may use any heuristics without affecting soundness, and emits a *certificate*. The *certificate* `c3_sandwich_certificate.txt` is a deterministic preorder serialization of $T(\tau_0)$ storing only the tree shape — for each internal node the split coordinate and the exact rational split point, for each leaf the pruning rule and its exact witness (the plank index for (P2)/(P4), the uncovered edge or the positive-area sign cell for (P3)); it stores no boxes. The *checker* `bb_certificate_check.py` (about one page, importing nothing from the searcher) reconstructs every box from the tree shape alone — so a box-arithmetic error in the searcher would surface here — and verifies (i) *tiling*: the stream is the preorder of a binary tree whose root is $[0,1]^6$ and each internal node splits one coordinate at a strictly interior point, the whole stream is consumed, the reconstruction stack ends empty, and the binary-tree identity (#leaves = #internal + 1) holds; and (ii) *prunes*: each leaf is re-validated against its recorded rule and witness by an independent re-implementation of the triangle geometry, of $w_0(\tau)$, of exact polygon clipping, and of (P1)–(P4). It prints `CERTIFICATE VALID` only if all checks pass.

The verified run has input $\tau_0 = (\frac{13}{25}, \frac{1}{2}, \frac{1}{2})$, threshold $w_0(\tau_0) = \frac{2}{25}$ (recomputed by the checker and matched against the header), and output: 812,650 internal nodes, 812,651 leaves, all pruned, with leaf counts

$$\underbrace{286{,}024}_{\text{(P1)}} \quad \underbrace{53{,}907}_{\text{(P2)}} \quad \underbrace{428{,}672}_{\text{(P3) edge}} \quad \underbrace{2{,}354}_{\text{(P3) cell}} \quad \underbrace{41{,}694}_{\text{(P4)}}.$$

The genuinely computational content is the (P1) and (P3) leaves; the (P2) and (P4) leaves are discharged by Theorems 10.4 and 3.1, and all but 2,354 of the (P3) leaves close by the one-dimensional edge test of Lemma 10.6(a).

What carries the proof, and what does not. Only the *certificate* and the *checker* are load-bearing: the theorem holds because the checker confirms that the certificate is a valid finite covering certificate in the sense of Lemma 10.7. The searcher is *search evidence* only — how the certificate was found — and is irrelevant to correctness; one may discard it and re-verify the shipped certificate with the checker alone. The checker has been confirmed non-vacuous (it rejects certificates with a tampered header, a non-interior split, a false leaf rule, or a truncated tree).

Reproducibility. Environment: Python 3 (tested on 3.14) with the standard-library `fractions` and `hashlib` only — the checker needs no third-party package. To re-verify, run

```
python3 bb_certificate_check.py c3_sandwich_certificate.txt
```

whose expected final line is `CERTIFICATE VALID` (with the leaf counts above); it runs in a few minutes on one core and uses modest memory. The certificate (12,062,803 bytes) is regenerable deterministically by `c3_balanced_bb.py` (fixed DFS, midpoint bisection, first-widest split coordinate). Full SHA-256 (each split across two lines):

`searcher`	`b2468aa9acc91f4fbeec9b58afed706d3`
	`b865e7f0e49ecfa2c3804eae382f4cb`
`certificate`	`ead66ff2cfd547f8e2acd7c7bf5cce67d1`
	`14a0aa1146559fc5b065028de6b86c`
`checker`	`90a25ce8ea059abed4707b5314a164be4`
	`0298b1fe9fe42394d56a1c69c48eb1b`

The scripts and certificate are archived with the ancillary files under a tagged, immutable release (a DOI will be minted on acceptance). As independent corroboration, a second search on the relabelled triple $(\frac{1}{2}, \frac{1}{2}, \frac{13}{25})$ with split ratio $\frac{5}{11}$ produces a *different* tree (1,040,919 versus 1,625,301 nodes) with the same empty verdict; the authority for the theorem is nonetheless the independent checker, since a common-mode bug in the prune code would survive the two-tree test but not the checker's independent re-implementation.

- `median_rigidity_enumeration.py` — exhaustive symbolic enumeration: the median edge-tiling system has exactly three solutions (Lemma 5.2).
- `median_edgetilings_independent.py`, `median_edgetiling_types_crosscheck.py` — two independent exact re-checks of the same enumeration (no SymPy; and via the L/M/R trace formulas of Appendix A).
- `median_rigidity_centroid.py` — the centroid witness in Theorem 5.1.
- `cyclic_family_tight_rigidity.py` — Theorem 6.6: the identities of Lemma 6.7, positivity of the witnesses of Lemma 6.9, exact coverage checks at several family points, and the general-τ solutions of Proposition A.1.
- `role_patterns_classification.py` — Theorem 6.11: exact classification of all 216 role assignments.
- `bang_Nc_nogo_chord.py` — Proposition 9.1: exact segment tightness of the chord lemma.
- `transport_defect.py` — §7: the facet loop marginals (Theorem 7.5), exact concurrence defects, the frozen-weight identities (Theorem 7.15), and the sandwich example (Remark 7.20).
- `defect_duality.py` — §7 and §8: the facet dual certificate, the wedge-certificate algebra of Proposition 7.9 (including the sandwich dependency $c = (75, 76, 74)$), the sign-counting of Lemma 7.10, the centroid bound (Theorem 7.3), the cycle marginals of Theorem 8.2 for $2 \le d \le 7$, and the region algebra of Corollary 7.18.
- `simplex3_weighted_skeleton.py` — Theorem 8.1 (closed forms $a(\sigma), b(\sigma)$, uniqueness) and Proposition 8.4.
- `delta_eq_deltac.py` — Theorem 7.11 and Lemma 7.10: the counting lemma on 2928 exact external (q, a) samples, and $\delta = \delta_c$ with the exact point-in-triangle test on a sweep of 507 triples covering all three full-edge pattern classes.
- `supD_dual_hunt.py`, `supD_hunt_phase2.py` — the multi-knot dual machinery behind Remark 7.20 (float-guided search; every reported bound re-derived exactly), skeleton upper bounds, and the searches of Problem 10.12.
- `dual_certificates.py` — *self-contained* exact verifier (rational arithmetic, no optimization libraries) of the three archival certificates of Remark 7.20.
- `gardner_third_marginal.py` — the moment obstruction for pair-witness mixtures (any Gardner witness of two facet directions has third marginal δ_0), documenting a refuted route to Problem 10.12.
- `covering_constant.py` — Theorem 6.13 (the three polynomial identities, coverage and margins), the exact coverage oracle behind the experiments of §10, and Remark 7.22's symbolic skeleton bound.

- `c3_extreme_regime.py` — Theorem 10.4: the corner ranges and extents, the strict Case-A inequalities, and the Case-B spanning bounds (symbolic, plus exact oracle sanity checks).
- `c3_balanced_bb.py` — the searcher for Theorem 10.8; emits the certificate.
- `bb_certificate_check.py` — the *independent* certificate checker (no shared code) for Theorem 10.8: tiling, the binary-tree identity, and re-validation of all 812,651 leaves; the sole computer-assisted step of the paper.
- `r15_equality.py` — the equality clause of Theorem 10.8: exact minimum widths $\frac{31}{25}$, $\frac{31}{25}$, $\frac{5}{4}$ of proper two-plank coverings.

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